

第 1 章 Maximum Likelihood Estimation

1.1 Likelihood function and the ML principle

Let $f(y; \theta_0)$ denote the probability density function of the random variable y given θ_0 . Our objective is to estimate the true parameter vector θ_0 . The joint density of n iid observations of y is

$$f(y_1, \dots, y_n | \theta_0) = \prod_{i=1}^n f(y_i; \theta_0) \quad (1)$$

Now taking $f(y_i; \theta)$ as a function of θ given $y_1, \dots, y_n$ and write

$$L(\theta | y_1, \dots, y_n) = \prod_{i=1}^n f(y_i; \theta) \quad (2)$$

gives the **likelihood function**, the likelihood that the population parameter is θ given the observed sample.

1.1.1 ML principle

The **Maximum Likelihood (ML) principle** suggests that estimators of the unknown parameters are obtained by **maximizing $L(\theta)$ with respect to θ** . If the random variable y is discrete, $\hat{\theta}$ yields the value of θ that maximizes the probability of observing the sample that actually occurred.

Usually, $\hat{\theta}$ can be obtained by solving the likelihood equation

$$\frac{\partial \log L(\theta)}{\partial \theta} \Big|_{\hat{\theta}} = 0 \quad (3)$$

where $\log L(\theta | y_1, \dots, y_n) = \sum_{i=1}^n \log f(y_i; \theta)$. Occasionally the ML estimator is not unique and $\log L(\theta)$ may have only one global maximum, but multiple local maxima.

The **regularity conditions** are the following:

1. The first three derivatives of $\log f(y|\theta)$ with respect to θ are continuous and finite for almost all y and for all θ .
2. For all values of θ , $|\frac{\partial^3 \log f(y|\theta)}{\partial \theta_j \partial \theta_k \partial \theta_l}|$ is limited by a function that has finite expectations.
3. The domain of y does not depend on θ . **For example of the uniform distribution, since the density function depends on θ_0 , the theorem does not hold.**
4. θ is an interior point to the compact parameter space Θ .

1.1.2 Bartlett Identities

Define the **score vector** $S(\theta)$ and the **Hessian matrix** $H(\theta)$ as

$$S(\theta) = \frac{\partial \log L(\theta)}{\partial \theta} = \sum_{i=1}^n \frac{\partial \log f(y_i|\theta)}{\partial \theta} = \sum_{i=1}^n s_i(\theta) \quad (4)$$

$$H(\theta) = \frac{\partial^2 \log L(\theta)}{\partial \theta \partial \theta'} = \sum_{i=1}^n \frac{\partial^2 \log f(y_i|\theta)}{\partial \theta \partial \theta'} = \sum_{i=1}^n H_i(\theta) \quad (5)$$

- The **First Bartlett identity** states that $E[s_i(\theta_0)] = 0$. Hence, $E[S(\theta_0)] = 0$.
- The **Second Bartlett identity** states that $\text{Var}[s_i(\theta_0)] = -E[H_i(\theta_0)]$.

证明.

$$\text{Var} \left[\frac{\partial \log f(y_i|\theta)}{\partial \theta} \right] = -E \left[\frac{\partial^2 \log f(y_i|\theta_0)}{\partial \theta \partial \theta'} \right]$$

where y_i is a random variable. □

$\text{Var}[s_i(\theta)] = E[s_i(\theta_0)s_i(\theta_0)']$ defines the **Fisher's information matrix** denoted as $\mathcal{I}(\theta_0)$. Hence, the result $\text{Var}[s_i(\theta)] = E[s_i(\theta_0)s_i(\theta_0)'] = -E[H_i(\theta_0)]$ is also called the **information matrix identity**.

证明.

$$\begin{aligned} \text{Var}[s_i(\theta)] &= E \left[\underbrace{[s_i(\theta_0) - E[s_i(\theta_0)]]}_{=0, \text{ First Bartlett}} \times \underbrace{[s_i(\theta_0) - E[s_i(\theta_0)]]'}_{=0, \text{ First Bartlett}} \right] \\ &= E[s_i(\theta_0)s_i(\theta_0)'] \end{aligned}$$

□

1.2 Properties of MLE

Under the assumed regularity conditions of MLE possesses the following properties:

1. **Consistency** meaning $\text{plim} \hat{\theta} = \theta_0$
2. **Asymptotic normality** meaning $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta_0)^{-1})$. The goal is to estimate the $[\mathcal{I}(\theta_0)]^{-1}$ matrix (Fisher's information matrix) $\rightarrow$ need to go to a known distribution (normal).
3. **Asymptotic efficiency** meaning that if $\tilde{\theta}$ is a regular consistent asymptotically normal estimator such that $\sqrt{n}(\tilde{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \Omega)$, then $\Omega - [\mathcal{I}(\theta_0)]^{-1}$ is positive semi-definite, i.e., under these RC, the MLE asymptotically achieves the **Cramer-Rao** lower bound which is given by $[\mathcal{I}(\theta_0)]^{-1}$. Information matrix is smaller than Ω in the matrix sense, and the information matrix is the lower bound.
4. **Invariance** meaning that if $c(\theta)$ is a continuous and continuously differentiable one-to-one function, the MLE of $\gamma = c(\theta_0)$ is $c(\hat{\theta})$.

Now, don't forget that θ_0 is not known, so we want to estimate the information matrix.

1.2.1 Estimators of the information matrix

There are three commonly used estimators of $\mathcal{I}(\theta_0)$:

- **Expected Information:** If the form of the expected values of the second derivatives of the log-likelihood function is known, then we can evaluate $\mathcal{I}(\theta)$ at $\hat{\theta}$. *This rarely happens in practice.*
- **Observed information:** Simply use $-\hat{H}(\hat{\theta})$.

证明.

$$\begin{aligned}
 \mathcal{I}(\theta_0) &= \text{Var}[s_i(\theta_0)] = -E[H(\theta_0)] \\
 &= \text{Var}\left[\frac{\partial \log f(y_i|\theta_0)}{\partial \theta_0}\right] \\
 &= -E\left[\frac{\partial^2 \log f(y_i|\theta_0)}{\partial \theta \partial \theta'}\right] \\
 &\stackrel{\text{avg.}}{=} \frac{1}{N} \sum_{i=1}^n \left(\frac{\partial^2 \log f(y_i|\theta_0)}{\partial \theta \partial \theta'}\right) \\
 &= -\hat{H}(\hat{\theta})
 \end{aligned}$$

□

- **Outer Product of the Gradient (OPG):** Because of the information matrix identity, we can also use $n^{-1} \sum_{i=1}^n s_i(\hat{\theta}) s_i(\hat{\theta})'$. *The OPG is notorious for its poor performance with finite samples.*

证明.

$$\begin{aligned}
 \text{Var}[s_i(\theta_0)] &= E[s_i(\theta_0) s_i(\theta_0)'] \\
 &= \frac{1}{N} \sum_{i=1}^N s_i(\theta_0) s_i(\theta_0)' \\
 &= \frac{1}{N} \sum_{i=1}^N s_i(\hat{\theta}) s_i(\hat{\theta})'
 \end{aligned}$$

□

1.3 Regressors

Suppose the joint distribution of y and x depends on α_0 , giving

$$f(y, x|\alpha_0) = f(y|x, \alpha_0)g(x|\alpha_0) \tag{6}$$

$$f(y|x, \alpha_0) = \frac{f(y, x, \alpha_0)}{g(x, \alpha_0)} \tag{7}$$

$$\underbrace{f(y, x, \alpha_0)}_{\text{joint}} = \underbrace{f(y|x, \alpha_0)}_{\text{conditional}} \underbrace{g(x, \alpha_0)}_{\text{marginal}} \tag{8}$$

and suppose α_0 can be divided into θ_0 and δ_0 , so that with exogeneity of x , $f(y, x|\alpha_0) = f(y_i|x_i, \theta_0)g(x_i|\delta_0)$. For an iid sample $\{(y_i, x_i)\}_{i=1}^n$,

$$\log L(\theta, \delta|y_i, x_i) = \sum_{i=1}^n \log f(y_i, x_i|\alpha) = \sum_{i=1}^n \log f(y_i|x_i, \theta) + \sum_{i=1}^n \log g(x_i|\delta) \quad (9)$$

We are under the assumption that we are only interested in θ_0 . Now this assumption that we don't care about δ makes our lives easier because

$$\frac{\partial L(\theta, \delta|y_i, x_i)}{\partial \theta} \Big|_{\hat{\theta}} = 0 \Leftrightarrow \frac{\partial \sum_{i=1}^n \log f(y_i|x_i, \theta_i)}{\partial \theta} \Big|_{\hat{\theta}} = 0$$

The part of $g(x, \delta)$ can be ignored now, only focus on the conditional density function. However, now $E[-H_i(\theta_0)] = \mathcal{I}(\theta_0)$.

1.4 Interpretation and inference in misspecified models

If the likelihood function is **misspecified**, the MLE is generally **inconsistent** for the parameters of interest. However, under the general conditions, $\text{plim} \hat{\theta} = \theta^*$, where the **pseudo-true value** θ^* minimizes the Kullback-Leibler divergence where $f_0(y)$ is the true distribution of data.

定义 1.4.1. Kullback-Leibner divergence, also called relative entropy, is a measure of how one probability distribution is different from a second, reference probability distribution.

This means that the MLE leads to the best approximation in the Kullback-Leibner sense to the true density. However, because Fisher's information matrix identity does not hold, the asymptotic covariance matrix is given by:

$$A^{-1}BA^{-1} \quad A = E[-H_i(\theta^*)] \quad B = E[s_i(\theta^*)s_i(\theta^*)'] \quad (10)$$

$f(y|\theta^*)$ is the best approximation given the wrong space but is still wrong.

1.5 Hypothesis testing

Considering a general set of restrictions to be tested

$$H_0 : h(\theta_0) = 0 \quad (11)$$

where

- θ : vector of parameters in the model and the true value is given by θ_0 .
- $h(\theta)$ is a $d \times 1$ vector of restrictions.
- $\hat{\theta}$ is the unrestricted MLE where its value maximizes $L(\theta)$.
- $\tilde{\theta}$ is the restricted MLE where its value maximizes $L(\theta)$ and satisfies $h(\theta) = 0$.

1.5.1 The Wald Test

Compares $h(\hat{\theta})$ with 0 since $h(\tilde{\theta}) = 0$. The test statistic is

$$\mathcal{W} = n \times h(\hat{\theta})' \left[G(\hat{\theta})' [\mathcal{I}(\hat{\theta})]^{-1} G(\hat{\theta}) \right]^{-1} h(\hat{\theta}) \quad (12)$$

where $G(\theta) = \frac{\partial h(\theta)}{\partial \theta}$.

Under the null hypothesis,

$$\mathcal{W} \xrightarrow{d} \chi^2(d) \quad (13)$$

Wald test is not invariant to how restrictions are formulated. Equivalent non-linear ($\beta_0/(1-\alpha_0) = 1$) and linear ($\beta_0 + \alpha_0 - 1 = 0$) restriction may lead to different values of $\mathcal{W}$.

1.5.2 The Likelihood Ratio Test

Compare $L(\hat{\theta})$ and $L(\tilde{\theta})$. If $h(\theta_0) = 0$ then $L(\hat{\theta})$ should be close to $L(\tilde{\theta})$. The test is base on the likelihood ratio $\lambda = \frac{\mathcal{L}(\tilde{\theta})}{\mathcal{L}(\hat{\theta})}$ with the test statistic

$$\mathcal{LR} = -2 \log(\lambda) = 2\{\log \mathcal{L}(\hat{\theta}) - \log \mathcal{L}(\tilde{\theta})\} \quad (14)$$

We may encounter convergence problem when maximizing.

Under the null hypothesis,

$$\mathcal{LR} \xrightarrow{d} \chi^2(d) \quad (15)$$

1.5.3 The Langrage Multiplier or Score Test

Compare $S(\tilde{\theta})$ with 0 since $S(\hat{\theta}) = 0$. The test statistic is

$$\mathcal{LM} = S(\tilde{\theta})' [\mathcal{I}(\tilde{\theta})]^{-1} S(\tilde{\theta}) \frac{1}{n} \quad (16)$$

Under the null hypothesis,

$$\mathcal{LM} \xrightarrow{d} \chi^2(d) \quad (17)$$

The LM test works best in real life application. It is under weaker regulated conditions but it's still difficult to compute the $S(\theta)$ function.

第 2 章 Binary Choice Models

2.1 Linear probability model

In many applications, the variate of interest is binary, i.e., takes values 0 and 1. Let's denote $X = (X_1, \dots, X_k)'$ and the objective of the regression model is to estimate $E[Y|X]$ where $E[Y|X] = P(Y = 1|X)$.

In the **linear probability model** we assume that

$$P(Y = 1|X) = \beta_1 + \beta_2 X_2 + \dots + \beta_k X_k \quad (18)$$

thus the **interpretation of β_j** is the **change in the probability of success when x_j changes**.

$$\frac{\partial P(Y = 1|X)}{\partial X_j} = \beta_j, \quad j = 2, \dots, k \quad (19)$$

and the **predicted Y** is the **predicted probability of success**.

The linear probability model is estimated using **OLS**. But there is the potential problem that the fitted values can be outside $[0, 1]$. Even without predictions outside $[0, 1]$, we may estimate effects that imply a change in x changes the probability by more than +1 or -1. **$\hat{\beta}$ too big that would not make sense**. Also, this model **violates the assumption of homoskedasticity**, therefore we should use the Eicker-Huber-White **robust standard errors** to make inference. **Homoskedasticity states that $\text{Var}[Y|X]$ does not vary with x but here it necessarily depends on x .**

证明.

$$\text{Var}[Y|X] = \underbrace{E[Y^2|X]}_{(1)} - E[Y|X]^2$$

(1):

$$E[Y^2|X] = 1^2 \times P(Y = 1|X) + 0^2 \times P(Y = 0|X) = P(Y = 1|X)$$

Now we have,

$$\text{Var}[Y|X] = P(Y = 1|X) - P(Y = 1|X)^2 = \underbrace{P(Y = 1|X)}_{\beta_1 + \beta_2 x_2 + \dots + \beta_k x_k} [1 - P(Y = 1|X)]$$

So we see that the variance depends on the regressors. □

2.2 Index models for binary response

The alternative to the linear model is to assume that $E[Y|X] = P(Y = 1|X) = G(X'\beta_0)$, where $G(\cdot)$ is bounded between 0 and 1. Thus, we have,

$$Y = \begin{cases} 1 & \text{with probability } G(X'\beta_0) \\ 0 & \text{with probability } 1 - G(X'\beta_0) \end{cases} \quad (20)$$

with the log-likelihood function of

$$\log\{L(\beta)\} = \sum_{i=1}^n Y_i \log(G(X'_i\beta)) + \sum_{i=0}^n (1 - Y_i) \log(1 - G(X'_i\beta)) \quad (21)$$

The **MLE** should be used to find the estimators, $\hat{\beta}_{ML}$.

Note that it is a system of non-linear equations to be solved to find $\hat{\beta}_{ML}$ thus we have to resort to numerical methods. **There is no closed form solution for this estimator.** **Consistency and asymptotic normality** follow the general results described for MLE under regularity conditions. Essentially, the **main requirement for consistency** is $E[Y|X] = P(Y = 1|X) = G(X'\beta_0)$.

In a correctly specified model, $\hat{\beta}$ is consistent and asymptotically normally distributed with variance-covariance matrix $[\mathcal{I}(\beta_0)]^{-1}$,

$$\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} \mathcal{N}(0, [\mathcal{I}(\beta_0)]^{-1})$$

Inference is done using the Wald, Likelihood Ratio, and Lagrange multiplier statistics as usual.

2.2.1 $G(X'\beta_0)$ Choices

The most popular choices are

- The **Logit Model**

$$G(X'\beta_0) = \frac{\exp(X'\beta_0)}{1 + \exp(X'\beta_0)} \quad (22)$$

- The **Probit Model**

$$G(X'\beta_0) = \Phi(X'\beta_0) \quad (23)$$

where $\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z \exp(-\frac{u^2}{2}) du$ is the Standard Normal Distribution function.

Both models are non-linear and require **MLE**. Other possible models are

- The **Log-Weibull Distribution**

$$G(X'\beta_0) = \exp(-\exp(X'\beta_0))$$

- The **Complementary log-log Model**, Gompertz Distribution

$$G(X'\beta_0) = 1 - \exp(-\exp(X'\beta_0))$$

•

$$G(X'\beta_0) = \Phi(X'\beta_0)^\tau$$

where τ needs to be estimated.

•

$$G(X'\beta_0) = 1 - \exp(-\omega \exp(X'\beta_0))^{-\frac{1}{\omega}}$$

where ω needs to be estimated. Note that for $\omega = 1$ we have the Logit model, and $\lim_{\omega \rightarrow 0} G(X'\beta_0) = 1 - \exp(-\exp(X'\beta_0))$.

Estimating additional parameters of the distribution allows the data to tell us the true model which is a major benefit.

In statistics, a more common interpretation of the coefficients is in terms of **marginal effects** on the odds ratio rather than on the probability. $\frac{P(Y=1|X)}{(1-P(Y=1|X))}$ measures the **probability of yes relative to no** and is called the **odds ratio** or relative risk. For the logit model, the log-odds ratio is linear in the regressors.

证明.

$$\begin{aligned} P(Y = 1|X) &= \frac{\exp(X'\beta_0)}{1 + \exp(X'\beta_0)} \Rightarrow \\ \frac{P(Y = 1|X)}{1 - P(Y = 1|X)} &= \frac{\frac{\exp(X'\beta_0)}{1 + \exp(X'\beta_0)}}{1 - \frac{\exp(X'\beta_0)}{1 + \exp(X'\beta_0)}} = \exp(X'\beta_0) \Rightarrow \\ \log\left(\frac{P(Y = 1|X)}{1 - P(Y = 1|X)}\right) &= X'\beta_0 \end{aligned}$$

□

2.2.2 Latent Variable Threshold (LVT) Model

Define a latent variable

$$Y^* = X'\beta_0 + \epsilon_t \quad (24)$$

where Y^* is unobserved, thus a **latent variable**. Further assume that ϵ independent of X , $E[\epsilon] = 0$ and $\text{Var}(\epsilon) = \sigma^2$ with a distribution function of $F(\cdot)$. The observation of Y is given by

$$Y = \begin{cases} 1 & \text{if } Y^* > \lambda \\ 0 & \text{if } Y^* \leq \lambda \end{cases} \quad (25)$$

where λ is the threshold.

The probability of choosing the option is thus given by,

$$\begin{aligned}
 P(Y = 1|X) &= P(Y^* > \lambda|X) = P(X'\beta_0 + \epsilon > \lambda|X) \\
 &= P(\epsilon > -X'\beta_0 + \lambda|X) \\
 &= 1 - P(\epsilon \leq -X'\beta_0 + \lambda|X) \\
 &\text{independence between } \epsilon, X \\
 &= 1 - \underbrace{F(-X'\beta_0 + \lambda)}_{\text{cumulative distribution function}} \\
 &= G(X'\beta_0)
 \end{aligned}$$

with $G(z) = 1 - F(-z + \lambda)$.

First identification problem:

$$P(Y = 1|X) = 1 - F(-X'\beta_0 + \lambda) \quad (26)$$

Notice that if $X_1 = 1$ meaning that **there is an intercept in the model**, it is not possible to estimate $-\beta_1$ and λ at the intercept separately. Solution is to **set $\lambda = 0$** . If $\lambda = 0$, and ϵ has a symmetric distribution around 0, then $G(z) = F(z)$ as in this case $1 - F(-z) = F(z)$.

Second identification problem:

$$\frac{Y^*}{a} = X'\beta_0^* + \frac{\epsilon}{a} \quad (27)$$

If we divide Y^* from (24) by a constant a and $a > 0$, notice that the definition of the observable variable Y (25) does not change. **Changing the scale $\rightarrow$ changing of variance but the Y does not change $\Rightarrow$ we cannot identify the scale of Y^*** . This implies that **we cannot identify the variance of ϵ** . **Error term variance not observed and cannot be identified and estimated**. For given β_0^* , value of β_0 depends on a , meaning β_0 is identified up to a scale factor. The solution is to **normalize distribution of ϵ** by fixing σ^2 at a given number $\Rightarrow$ assume σ^2 is known. **Normally set to 1 in practice**.

2.2.3 Random Utility Models

Suppose an individual has to choose between alternatives a and b with utilities U^a and U^b which are unobservable. But the characteristics of the observations are observable thus we have,

$$\begin{aligned}
 U^a &= X^a\beta_a + u_a \\
 U^b &= X^b\beta_b + u_b
 \end{aligned}$$

We also observe the chosen alternative, say a , which we indicate this choice as $Y = 1$. Then we know that,

$$\begin{aligned} P(Y = 1|X) &= P(U^a > U^b|X) = P(X'\beta_a + u_a > X'\beta_b + u_b) \\ &= P(u_a - u_b > X'(\beta_b - \beta_a)|X) \\ &= P(\epsilon > -X'\beta_0|X) = 1 - F(-X'\beta_0) \end{aligned}$$

with $\epsilon = u_a - u_b$ and $\beta_0 = \beta_a - \beta_b$.

2.3 Marginal effects and average partial effects

It is important to note that the parameters of the model, like those of any nonlinear regression model, are not necessarily the marginal effects. In general, the partial effect associated with variable k , X_{ik} is given by,

$$\Delta E[Y_i|X_i] \equiv \Delta P(Y_i = 1|X_i) \simeq \frac{\partial E[Y_i|X_i]}{\partial X_{ik}} \Delta X_{ik} = \frac{\partial G(X_i'\beta_0)}{\partial X_{ik}} \Delta X_{ik} \quad (28)$$

Let $\omega = X_i'\beta_0 = \beta_1 X_{i2} + \dots + \beta_K X_{iK}$, then,

$$\frac{\partial E[Y_i|X_i]}{\partial X_{ik}} = \frac{\partial G(\omega)}{\partial(\omega)} \frac{\partial(\omega)}{\partial X_{ik}} = g(\omega)\beta_k = g(X_i'\beta_0)\beta_k \quad (29)$$

where $g(\omega) = \frac{\partial G(\omega)}{\partial \omega}$. Thus, we need to evaluate

$$\Delta e[Y_i|X_i'\beta_{0i}] \equiv \Delta P(Y_i = 1|X_i) \simeq g(X_i'\beta_0)\beta_k \Delta X_{ik} \quad (30)$$

Now, to **compute marginal effects** of variable X_k assuming that X_{ik} is **continuous**, we have options

- Marginal effect at a **particular value** $X_i = X_0$

$$g(X_0'\hat{\beta}_{ML})\hat{\beta}_{ML,i}\Delta X_{ik}$$

The most common choices of X_0 are the vector of sample means. This does not make sense if some of the regressors are dummy variables. Or the vector of sample medians.

- Alternative is to compute **Average Partial Effects** which is the sample average of the individual marginal effects,

$$\frac{1}{n} \sum_{i=1}^n g(X_i'\hat{\beta}_{ML})\hat{\beta}_{ML,k}\Delta X_{ik}$$

The options to compute the marginal effects on a **dummy variable** d , we have the same options

- Marginal effect at a **particular value** $X_i = X_0$

$$P(Y_i = 1|\widehat{X_0}, d_i = 1) - P(Y_i = 1|\widehat{X_0}, d_i = 0)$$

Note that here X_0 denotes all variables in the model at the particular value X_0 except d .

- Average partial effect is thus given by

$$\frac{1}{n} \sum_{i=1}^n P(Y_i = 1|\widehat{X_i}, d_i = 1) - \frac{1}{n} \sum_{i=1}^n P(Y_i = 1|\widehat{X_i}, d_i = 0)$$

Note that here X_i denotes all variables in the model except d .

2.4 Simple specification tests

2.4.1 Misspecification

If $G(\cdot)$ is **misspecified**, then $\hat{\beta}_{ML}$ is **inconsistent**. Perform **RESET-type test** by testing $H_0 : \delta_1 = \delta_2 = 0$ in the model

$$E[Y_i|X_i] = G(X_i'\beta_0 + \delta_1(X_i'\hat{\beta}_{ML})^2 + \delta_2(X_i'\hat{\beta}_{ML})^3), \quad i = 1, \dots, n$$

This is a normality test in the probit if $\epsilon_i \sim \mathcal{N}(0, 1) \rightarrow f(\cdot) = \Phi(\cdot)$. $E[Y_i|X_i] = X_i'\beta$ estimated by OLS and the estimator $\hat{\beta}$ and the model $Y_i = X_i'\beta + \delta_0(X_i'\beta)^2 + \delta_1(X_i'\beta)^3$ and test $H_0 : \delta_0 = 0$ and $\delta_1 = 0$, if we reject $H_0 \rightarrow$ model is misspecified.

The RESET test can be tested against more general parametric specifications which include additional shape parameters. Consider $G(\cdot) = \Phi(\cdot)^\tau$, use the score statistic to test $H_0 : \tau = 1$, Probit. Or, consider $G(\cdot) = 1 - (1 + \omega \exp(\cdot))^{-\frac{1}{\omega}}$, use the score statistic to test $H_0 : \omega = 1$, Logit.

2.4.2 Heteroskedasticity

In the LVT model, heteroskedasticity leads to misspecification of the conditional mean of Y .

证明. Proof of misspecification for LVT.

The goal is to show that $E[Y|X] \neq G(X'\beta_0)$. As previously shown, $G(a) = 1 - F(a)$.

$$P(Y = 1|X) = P(Y^* > 0|X) = P(X'\beta_0 + k h(Z'\gamma_0)\epsilon > 0|X)$$

$$= P\left(\epsilon > \frac{-X'\beta_0}{k h(Z'\gamma_0)} | X\right)$$

notice X' and condition on X , we can drop the condition

$$= P\left(\epsilon > \frac{-X'\beta_0}{k h(Z'\gamma_0)}\right)$$

$$= 1 - P\left(\epsilon \leq \frac{-X'\beta_0}{k h(Z'\gamma_0)}\right)$$

$$= 1 - F\left(\epsilon \leq \frac{-X'\beta_0}{k h(Z'\gamma_0)}\right)$$

$$= G\left(\epsilon \leq \frac{-X'\beta_0}{k h(Z'\gamma_0)}\right) \neq G(X'\beta_0) \rightarrow \text{misspecified}$$

□

Consider a latent variable,

$$Y^* = X'\beta_0 + k \times h(Z'\gamma_0)\epsilon \quad (31)$$

under the assumptions of ϵ independent of X , $E[\epsilon] = 0$ and $\text{Var}(\epsilon) = 1$ and distribution function $F(\cdot)$, Z are a vector function of X of size d and h any function with $h > 0$, $h(0) = 1$, $h'(0) \neq 0$.

证明. Proof of heteroskedasticity.

The idea is to prove that the variance of the error term is not constant but a function of X .

$$\begin{aligned}
 u &= k \times h(Z' \gamma_0) \epsilon \\
 e[u|X] &= E[k \times h(Z' \gamma_0) \epsilon | X] = k \times h(Z' \gamma_0) \underbrace{E[\epsilon | X]}_{=0} = 0 \\
 \text{Var}(u|X) &= \text{Var}(k \times h(Z' \gamma_0) \epsilon | X) = k^2 [h(Z' \gamma_0)]^2 \underbrace{\text{Var}(\epsilon | X)}_{=1} = k^2 [h(Z' \gamma_0)]^2 \rightarrow \text{Heteroskedasticity}
 \end{aligned}$$

□

We can construct a **LM test** based on the generalized residuals to test $H_0 : \gamma_0 = 0$ which implies **homoskedasticity**. The LM test statistic is calculated as

$$\xi_{LM} = \iota' S (S' S)^{-1} S' \iota \sim \chi^2(d)$$

This is asymptotically equivalent to a **Wald test** with $H_0 : \gamma_0 = 0$ in the model

$$E[Y_i | X_i] = G(X' \beta_0 + (X'_i \hat{\beta}_{ML}) Z'_i \gamma_0), \quad i = 1, \dots, n$$

2.4.3 Goodness of fit

Unlike the linear probability model, we cannot compute an R^2 to judge the goodness of fit. One possibility is to use the **pseudo R^2** based on the log likelihood. We can also look at the percent correctly predicted. [We can just use the percentage of correct predictions and we can adjust the threshold. This is more for machine learning when prediction matters.](#) For econometrics, the interpretation of β is more important, thus the specification test is more important.

第 3 章 Ordered Data and Count Data Models

3.1 Ordered data

In some problems, the variate of interest assumes more than two discrete outcomes, but these are inherently **ordered**. Ordered data is a generalization of the binary choice model but with an order. Zavoina and McElvey (1975) modeled ordered data using the following latent variable framework,

$$Y_i^* = X_i' \beta_0 + u_i, \quad Y_i = \begin{cases} 0 & Y_i^* \leq \mu_0 \\ 1 & \mu_0 < Y_i^* \leq \mu_1 \\ 2 & \mu_1 < Y_i^* \leq \mu_2 \\ \vdots & \vdots \\ J-1 & \mu_{J-2} < Y_i^* \leq \mu_{J-1} \\ J & \mu_{J-1} < Y_i^* \end{cases} \quad (32)$$

where the threshold parameters are such that $0 = \mu_0 < \mu_1 < \dots < \mu_{J-1}$ and Y_i^* is the latent variable.

If the distribution of u_i is specified, the unknown parameters β and $\mu_2, \dots, \mu_{J-1}$ can be estimated by **ML**. Assume that u_i has distribution function $F(\cdot)$ and is independent of X_i , notice that,

$$\begin{aligned} P_0(X_i, \beta_0) &= P(Y_i = 0|X_i) = F(-X_i' \beta_0) \\ P_1(X_i, \beta_0) &= P(Y_i = 1|X_i) = F(\mu_1 - X_i' \beta_0) - F(-X_i' \beta_0) \\ &\vdots \\ P_j(X_i, \beta_0) &= P(Y_i = j|X_i) = F(\mu_j - X_i' \beta_0) - F(\mu_{j-1} - X_i' \beta_0) \\ &\vdots \\ P_J(X_i, \beta_0) &= P(Y_i = J|X_i) = 1 - F(\mu_{J-1} - X_i' \beta_0) \end{aligned}$$

Therefore the log-likelihood function is,

$$\log L(\theta) = \sum_{i=1}^n \sum_{j=1}^J 1(Y_i = j) \log(P_j(X_i, \beta))$$

As in all discrete choice models, the variance of u_i is not identified. And the **ordered-probit and ordered-logit** are the most used special cases of the model.

Interpreting coefficients requires some care and is messy. But, the OP and OL models allow us to obtain the **sign of the partial effects** on $P(Y > j|X_i)$. For a continuous variable x_h for the OP model,

$$\frac{\partial P(Y_i > j|X_i)}{\partial x_h} = \beta_h \Phi(\mu_j - X_i' \beta)$$

If $\beta_h > 0$, **an increase in x_h increases the probability that Y_i is greater than any value j** . Of course, we can interpret the sign of the parameters in the latent variable model. **It is actually easier to interpret the β 's of the latent equation to avoid the complications of finding the marginal effect.**

A closely related model can be used for **grouped data**. In this case, the threshold parameters are the limits of the intervals. The main difference is that for $J > 0$, **the variance of u_i is identified** because the thresholds give information on the scale of u_i . **Only need to estimate the β 's because the μ 's are known.**

3.2 Count data models

In many applications, the variate of interest is the count of the number of occurrences of some event in a given period of time. These data have some very specific characteristics: discreteness, non-negative, and many zeros and a long right-hand tail.

In those conditions, the **standard linear models are not appropriate** because:

- The conditional expectation is necessarily non-negative. **The mean is always positive.**
- The data is intrinsically heteroskedastic.
- Do not allow the computation of the probability of events of interest.

3.2.1 The Poisson Regression Model

The **Poisson regression model** is defined by,

$$P(Y_i = j|X_i) = \frac{\exp\{-\lambda(X_i, \beta_0)\} \lambda(X_i, \beta_0)^j}{j!}, \quad j = 0, 1, 2, \dots \quad (33)$$

with

$$\begin{aligned} E[Y_i|X_i] &= \text{Var}(Y_i|X_i) = \lambda(X_i, \beta_0) \quad \text{equidispersion assumption} \\ \text{Var}(Y_i) &= E_x[\lambda(X_i, \beta_0)] + \text{Var}[\lambda(X_i, \beta_0)] \geq E_x[\lambda(X_i, \beta_0)] = E[Y_i] \end{aligned}$$

where in general the specification of $\lambda(X_i, \beta_0) = \exp(X_i' \beta_0)$ is adopted. **Because $\exp(X_i' \beta_0) > 0$** . Therefore,

$$\frac{\partial E[Y_i|X_i]}{\partial X_i} = \exp(X_i' \beta_0) \beta_0$$

and **ML** estimation of β_0 is straightforward.

证明. Proof of $\text{Var}(Y_i)$

$$\begin{aligned}
 \text{Var}(Y_i) & \text{ using law of total variance} \\
 & = \overbrace{\text{Var}_{X_i}(E[Y_i|X_i])}^{\geq 0} + E_{X_i}[\underbrace{E[Y_i|X_i]}_{\lambda(\cdot)}] \Rightarrow \\
 \text{Var}(Y_i) & \geq E_{X_i}[\text{Var}(Y_i|X_i)] = E_{X_i}[\lambda(X_i, \beta_0)] \\
 & = E_{X_i}[E[Y_i|X_i]] \quad \text{using the law of iterated expectations} \\
 & = E[Y_i]
 \end{aligned}$$

□

The log-likelihood function is given by,

$$\log L(\beta) = \sum_{i=1}^n [-\exp(X_i'\beta) + (X_i'\beta)Y_i - \log(Y_i!)]$$

The **Hessian is negative definite** for all X and β which facilitates the estimation and ensures the uniqueness of the maximum if it exists. The **MLE** has the usual properties,

$$\sqrt{n}(\hat{\beta}_{ML} - \beta_0) \xrightarrow{d} \mathcal{N}(0, E[\exp(X_i'\beta_0)X_iX_i']^{-1})$$

And as usual, **inference** can be performed using the **LR, W, and LM tests**.

3.2.2 Overdispersion

The Poisson model imposes equidispersion which is very restrictive.

定义 3.2.1. Equidispersion means the conditional expectation is equal to the variance.

$$E[Y_i|X_i] = \text{Var}(Y_i|X_i)$$

While **overdispersion** means that the variance is bigger than the conditional mean.

$$\text{Var}(Y_i|X_i) > E[Y_i|X_i]$$

There are many causes for overdispersion: measurement error, misspecification of the condition mean, neglected heterogeneity (random parameter variation). Focusing on the neglected heterogeneity issue, assuming

$$E[Y_i|X_i, \epsilon_i] = \exp(X_i'\beta_0 + \epsilon_i) \quad \epsilon_i \text{ can be any possibilities of error}$$

$$E[\exp(\epsilon_i)|X_i] = 1$$

$$\text{Var}(\exp(\epsilon_i)|X_i) = \sigma^2$$

In this case,

$$E[Y_i|X_i] = E[\lambda(X_i, \beta_0)|X_i] = E_\epsilon[\exp(X_i'\beta_0 + \epsilon_i)|X_i] = \exp(X_i'\beta_0)$$

Therefore, this sort of neglected heterogeneity **does not change the form of the conditional expectation of Y_i** .

定理 3.2.2. Gouriéroux, Monfort and Trognon (1984)

If $E[Y_i|X_i] = \lambda(X_i', \beta_0)$ **is correctly specified** and the likelihood function is constructed using a probability distribution which does not necessarily correspond to the true distribution of the data, but **belongs to the family of linear exponential distribution**, then the **Quasi-Maximum Likelihood** estimator is **consistent** for β_0 .

Wrong estimator $\rightarrow$ same exponential family $\rightarrow$ still consistent. The family of linear exponential distribution includes

- Poisson distribution
- Normal distribution with fixed variance $\rightarrow$ using continuous distribution to describe a discrete distribution.
- Binomial distribution with fixed number of trials
- Gamma distribution with fixed shape parameter

Inference is done using the results presented for the Quasi-Maximum Likelihood estimator. In particular, since the Poisson pseudo-MLE is consistent in presence of this sort of misspecification, valid inference can be based on

$$\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} \mathcal{N}(0, A^{-1}BA^{-1})$$

where

$$A = E[\exp(X_i'\beta_0)X_iX_i'] \quad B = E[(Y_i - \exp(X_i'\beta_0))^2 X_iX_i']$$

Notice that

$$\begin{aligned} \text{Var}(Y_i|X_i) &= E_{\epsilon_i}[\text{Var}(Y_i|X_i, \epsilon_i)] + \text{Var}(E_{\epsilon_i}[Y_i|X_i, \epsilon_i]) \quad \text{law of total variance} \\ &= \exp(X_i'\beta_0) + \sigma^2 \exp(2X_i'\beta_0) \\ &\geq E[Y_i|X_i] = \exp(X_i'\beta_0) \end{aligned}$$

This means that the **presence of overdispersion or underdispersion** can be tested with **LM test statistic**

$$\begin{aligned} H_0 : \sigma^2 = 0 &\rightarrow \text{equidispersion} \\ T &= \sum_{i=1}^n \frac{(Y_i - \exp(X_i'\hat{\beta}))^2 - Y_i}{\sqrt{2 \sum_{i=1}^n \exp(2X_i'\hat{\beta})}} \xrightarrow{d} \mathcal{N}(0, 1) \end{aligned}$$

Or a **RESET test** to test if the specification $E[Y_i|X_i] = \exp(X_i'\beta_0)$ is correct.

$$\begin{aligned} H_o : \gamma_0 = 0 &\rightarrow \text{correctly specified thus equidispersion} \\ E[Y_i|X_i] &= \exp(X_i'\beta_0 + \gamma_0(X_i'\beta_0)^2) \end{aligned}$$

3.2.3 Heterogeneity and the Negative Binomial Regression Model

The assumption that Y_i has a Poisson distribution conditional of X_i and ϵ_i with mean $\lambda_i = \exp(X_i' \beta_0 + \epsilon_i)$ leads to the **compound Poisson regression model**.

$$P(Y_i = j | X_i, \epsilon_i) = \frac{\exp[-\exp(X_i' \beta_0 + \epsilon_i)] \exp(X_i' \beta_0 + \epsilon_i)^j}{j!} \quad (34)$$

$$P(Y_i = j | X_i) = \int_{-\infty}^{+\infty} \frac{\exp[-\exp(X_i' \beta_0 + \epsilon_i)] \exp(X_i' \beta_0 + \epsilon_i)^j}{j!} g(\epsilon_i) d\epsilon_i \quad (35)$$

where $g(\epsilon_i)$ is the density function of ϵ_i and we assume that X_i and ϵ_i are independent. This model can be made operational in different ways:

- Pseudo maximum likelihood estimation → [Refer to previous section](#).
- Parametric estimation for specified $g(\epsilon_i)$
- Semiparametric estimation of β_0 and $g(\epsilon_i)$

Now focusing on the second method when **$g(\epsilon_i)$ is specified**. **MLE** can be obtained, but the estimator **may not be robust** to departures from the additional distributional assumptions. Assume that $\exp(\epsilon_i) \sim \Gamma(\sigma^{-2}, \sigma^2)$, then $P(Y_i = j | X_i)$ is given by the **Negative-Binomial model**, Cameron and Trivedi (1986). The Poisson model is obtained as a limiting case when $\sigma^2 \rightarrow 0$, but $H_0 : \sigma^2 = 0$ cannot be tested with a standard LR or Wald test.

If the model is **misspecified** but $E[Y_i | X_i] = \exp(X_i' \beta_0)$ is correct and σ^{-2} is fixed, the **negative-binomial Pseudo-MLE** estimator is **consistent** for β_0 . This follows the result from **Gourieroux, Monfort and Trognon (1984)** and the fact that the negative-binomial distribution with σ^{-2} fixed is a member of the family of linear exponential distributions.

引理 3.2.3. *Building on top of the original **Gourieroux, Monfort and Trognon (1984)***

If the model

- *is correct $\xrightarrow{MLE}$ consistent*
- *is incorrect but fix $\sigma^{-2} \xrightarrow{MLE} \hat{\beta}$ consistent*
- *is incorrect $\xrightarrow{MLE} \hat{\beta}, \sigma^2$ inconsistent*

The **score test** for $H_0 : \sigma^2 = 0$ is the overdispersion test discussed previously. Other parametric alternatives to the Poisson regression are available.

Now focusing on the third method with **semiparametric estimation** which assumes that ϵ has a discrete

distribution with Q support points $\alpha_1, \dots, \alpha_Q$ and corresponding possibilities $\pi_1, \dots, \pi_Q$, leading to

$$P(Y_i = j|X_i) = \sum_{q=1}^Q \frac{\exp[-\exp(X_i' \beta_0 + \alpha_q)] \exp(X_i' \beta_0 + \alpha_q)^j}{j!} \pi_q \quad (36)$$

For a given Q , estimation of $\beta, \alpha_1, \dots, \alpha_Q, \pi_1, \dots, \pi_{Q-1}$ can be performed by **ML**. This leads to a **consistent** estimator if Q is allowed to increase at an appropriate rate. In practice, the value of Q has to be chosen, for example using information criterion. But **inference is complicated** by the fact that the number of parameters is not fixed. **This method is nice because the big integral in (35) now becomes $\sum_{q=1}^Q \pi_q = 1$ since ϵ_i is discrete.**

3.2.4 Hurdle and Zero-Inflated Poisson Model

In some cases, the population may be contaminated by individuals for which $Y_i \equiv 0$. There are two ways to model this type of data

1. Zero-Inflated Poisson Model
2. Hurdle Model

In the **Zero-Inflated Poisson Model**, the zero outcome can arise from one of two regimes. In one regime, the outcome is always zero. In the other, the usual Poisson process is at work. Let Z_i be a Bernoulli random variable such that

$$Z_i = \begin{cases} 0 & \text{with } P(Z_i = 0|X_i) = p_i \\ 1 & \text{with } P(Z_i = 1|X_i) = 1 - p_i \end{cases}$$

Let $P(Y_i = j|X_i, Z_i = 1) = \pi_i(j; \beta_0)$, $j = 0, 1, \dots$ be the Poisson probability function. And let $P(Y_i = 0|X_i, Z_i = 0) = 1$. Note that,

$$\begin{aligned} P(Y_i = 0|X_i) & \text{ law of total probability} \\ &= P(Z_i = 0|X_i)P(Y_i = 0|X_i, Z_i = 0) + P(Z_i = 1|X_i)P(Y_i = 0|X_i, Z_i = 1) \\ &= P(Z_i = 0|X_i) + P(Z_i = 1|X_i)P(Y_i = 0|X_i, Z_i = 1) \\ &= p_i + (1 - p_i)\pi_i(0; \beta_0) \end{aligned}$$

For $j > 0$,

$$\begin{aligned} P(Y_i = j|X_i) & \text{ law of total probability} \\ &= \underbrace{P(Z_i = 0|X_i)}_{=p_i} \underbrace{P(Y_i = j|X_i, Z_i = 0)}_{=0 \text{ because all the mass are at 0 so cannot be } j} + \underbrace{P(Z_i = 1|X_i)}_{=(1-p_i)} \underbrace{P(Y_i = j|X_i, Z_i = 1)}_{\pi(j; \beta_0)} \\ &= (1 - p_i)\pi(j; \beta_0) \end{aligned}$$

Also notice that,

$$\begin{aligned}
 E[Y_i|X_i] &= \sum_{j=0}^{\infty} jP(Y_i = j|X_i) = 0 \times P(Y_i = 0|X_i) + \sum_{j=1}^{\infty} jP(Y_i = j|X_i) \\
 &= \sum_{j=1}^{\infty} j(1 - p_i)\pi(j; \beta_0) \\
 &= (1 - p_i) \underbrace{\sum_{j=1}^{\infty} j\pi(j; \beta_0)}_{\exp(X_i'\beta_0) \text{ mean of the Poisson random variable}}
 \end{aligned}$$

Therefore, the standard **pseudo maximum likelihood does not hold here if p_i depends on X_i** . The log-likelihood function this model is written as

$$\log L(\beta) = \sum_{i=1}^n \log\{[p_i + (1 - p_i)\pi_i(0; \beta)]^{1(Y_i=0)} \times [(1 - p_i)\pi_i(Y_i; \beta)]^{1(Y_i>0)}\} \quad (37)$$

The **Hurdle Model** is a different extension of the basic count data model is obtained by letting the zero and positive observations be generated by different mechanisms. A binary probability model determines whether a zero or a nonzero outcome occurs, then in the latter case, we observe always a positive integer $1, 2, \dots$. Consider a Bernoulli random variable

$$W_i = \begin{cases} 1 & \text{with } P(W_i = 1|X_i) = 1 - q_i \\ 0 & \text{with } P(W_i = 0|X_i) = q_i \end{cases}$$

where q_i may depend on X_i .

$$P(Y_i = 0|X_i, W_i = 0) = 1$$

$$P(Y_i = 0|X_i, W_i = 1) = 0$$

$$P(Y_i = j|X_i, W_i = 1) = \pi^*(j; \beta_0) \quad j = 1, 2, \dots$$

As the model only takes strictly positive values, $\pi^*(j; \beta_0)$ cannot be Poisson. In this case,

$$\begin{aligned}
 P(Y_i = 0|X_i) &= P(W_i = 0|X_i)P(Y_i = 0|X_i, W_i = 0) + P(W_i = 1|X_i)P(Y_i = 0|X_i, W_i = 1) \\
 &= q_i \times 1 + (1 - q_i) \times 0 = q_i
 \end{aligned}$$

For $j = 1, 2, \dots$

$$\begin{aligned}
 P(Y_i = j|X_i) &= P(W_i = 0|X_i)P(Y_i = j|X_i, W_i = 0) + P(W_i = 1|X_i)P(Y_i = j|X_i, W_i = 1) \\
 &= P(W_i = 1|X_i)P(Y_i = j|X_i, W_i = 1) \\
 &= q_i \times 0 + (1 - q_i)\pi^*(j; \beta_0)
 \end{aligned}$$

In this case, we have

$$\begin{aligned}
 E[Y_i|X_i] &= \sum_{j=0}^{\infty} jP(Y_i = j|X_i) = 0 \times P(Y_i = 0|X_i) + \sum_{j=1}^{\infty} jP(Y_i = j|X_i) \\
 &= \sum_{j=1}^{\infty} j(1 - p_i)\pi^*(j; \beta_0) \\
 &= (1 - p_i) \underbrace{\sum_{j=1}^{\infty} j\pi^*(j; \beta_0)}_{\text{mean of } \pi^*(j; \beta_0) \text{ cannot be } \exp(X_i' \beta_0)}
 \end{aligned}$$

Again, the **standard pseudo maximum likelihood result does not hold here**. Then, the log-likelihood function has the form

$$\log L(\beta) = \sum_{i=1}^n \{1(Y_i = 0)(\log q_i) + 1(Y_i > 0) \log(1 - q_i) + 1(Y_i > 0) \log[\pi^*(Y_i; \beta)]\} \quad (38)$$

Notice that this function is separable meaning the **correlated unobserved heterogeneity can be allowed for** and integrated out numerically. Can model q_i 's separately, $q_i = \frac{\exp(X_i' \gamma_0)}{1 + \exp(X_i' \gamma_0)}$ with this then we can estimate the β 's.

Usually, $\pi^*(j; \beta_0)$ is specified as a truncated Poisson of the form

$$\pi^*(j; \beta_0) = \frac{\exp(-\lambda_i) \lambda_i^j}{(1 - \exp(-\lambda_i))j!}, \quad j > 0$$

with $\lambda_i = \exp(X_i' \beta_0)$. However, there is no real truncation in this model, so an equally valid specification is

$$\pi^*(j; \beta_0) = \frac{\exp(-\lambda_i) \lambda_i^{j-1}}{(j-1)!}, \quad j > 0$$

When the **truncated Poisson specification is used**, we can test the **null of no hurdle** through

$$\begin{aligned}
 H_0 : \beta_0 &= \gamma_0 \\
 q_i &= \exp(-\exp(X_i' \gamma_0))
 \end{aligned}$$

In any case, **consistency depends on the distributional assumptions**.

第 4 章 Limited Dependent Variable Models

In some applications, the domain of the variate of interest is limited. In application, the bounds of the domain of Y often have to be taken into account. Traditionally, there are three forms of limited dependent variable models:

1. Truncated data
2. Censored data
3. Sample selection which is also called incidental truncation

4.1 Truncated data

定义 4.1.1. A sample Y is said to come from a **truncated distribution** when it is not possible to obtain observations from part of the domain of Y .

If Y is a **continuous random variable** with pdf $f(y)$ and a is a constant, we have **left-truncation at a** if the sample is drawn from the conditional distribution

$$f(y|Y > a) = \frac{f(y)}{P(Y > a)}$$

We have also truncation if the regressors are also not observed when $Y < a$. For a **discrete random variable** with support on $0, 1, \dots, \infty$, we have **left-truncation at a** if the sample is drawn from

$$P(Y = j|Y > a) = \frac{P(Y = j)}{1 - \sum_{k=0}^{\lfloor a \rfloor} P(Y = k)'}.$$

Notice that truncation is only problematic if it depends on Y or on another variable that is not independent of Y and that is not used as a regressor. Under certain conditions, it is possible to perform inference about the entire population using truncated samples. The problem is that this inference tends to be **sensitive to distributional assumptions**.

Example

Normal distribution with mean μ and variance σ^2

$$E[Y_i|Y_i > a] = \mu + \underbrace{\sigma \frac{\phi(\frac{1-\mu}{\sigma})}{1 - \Phi(\frac{a-\mu}{\sigma})}}_{\text{extra term}}$$

If you try to make inference on μ , it would be biased because of this extra term.

For the case of count data with $E[Y_i|X_i] = \exp(X_i'\beta_0)$

$$E[Y_i|X_i, Y_i > a] = \frac{\exp(X_i'\beta_0)}{1 - \underbrace{\sum_{k=0}^{\lfloor a \rfloor} P(Y_i = k|X_i)}_{\text{notice here}}}$$

The result depends on the probability function of the data.

In both of the examples, the expectation for the truncated data **depends on the shape of the conditional distribution**. Because of this, the estimation is often performed by **maximum likelihood**. Assume we know the distribution of the data.

4.1.1 Truncated normal regression model

The **Truncated normal regression model** is defined by

$$Y_i^* = X_i'\beta_0 + u_i^*, \quad u_i^* \sim \mathcal{N}(0, \sigma^2)$$

$$(Y_i, X_i) = \begin{cases} (Y_i^*, X_i) & \text{if } Y_i^* > 0 \\ \text{non-observable} & \text{if } Y_i^* \leq 0 \end{cases}$$

This implies that

$$E[Y_i|X_i, Y_i > 0] = X_i'\beta_0 + \sigma \frac{\phi(\frac{-X_i'\beta_0}{\sigma})}{1 - \Phi(\frac{-X_i'\beta_0}{\sigma})}$$

Again with the extra term, the OLS assumption of $E[Y_i|X_i] = X_i'\beta_0$ is violated $\rightarrow$ use MLE. The parameters of interest can be estimated from the likelihood function

$$L(\beta, \sigma) = \prod_{i=1}^n \frac{1}{\sigma} \frac{\phi(\frac{Y_i - X_i'\beta}{\sigma})}{1 - \Phi(\frac{-X_i'\beta_0}{\sigma})}$$

4.2 Censored data

Censoring is a problem of partial observability. For example, if Y^* is a random variable and a is a constant, we have **left-censoring at a** if we can only observe

$$Y = \max\{Y^*, a\} = \begin{cases} a & \text{if } Y^* \leq a \\ Y^* & \text{if } Y^* > a \end{cases}$$

We can also have **right truncation**, in this case, $Y = \min\{Y^*, a\}$.

Focusing on left-censored data. If Y^* is a **discrete random variable** with support on $0, 1, \dots, \infty$, the probability function of Y is given by

$$P(Y = j) = \begin{cases} 0 & \text{if } j < a \\ P(Y^* \leq a) & \text{if } j = a \\ P(Y^* = j) & \text{if } j > a \end{cases}$$

If Y^* is a **continuous random variable** with pdf $f(y^*)$, then Y has the **mixed distribution**

$$P(Y \leq k) = \begin{cases} 0 & \text{if } k < a \\ P(Y^* \leq k) & \text{if } k \geq a \end{cases}$$

It is assumed that the **regressors are fully observed**, even for cases with censored Y^* .

4.2.1 Tobit model

The standard regression model for continuous censored data is the **Tobit** which was proposed by Tobin (1958). The model is defined by

$$Y_i^* = X_i' \beta_0 + u_i^*, \quad u_i^* \sim \mathcal{N}(0, \sigma^2)$$

$$Y_i = \begin{cases} 0 & \text{if } Y_i^* \leq 0 \\ Y_i^* & \text{if } Y_i^* > 0 \end{cases}$$

The model is appropriate if we are interested in **making inference about Y_i^* not Y_i** . In this case,

$$E[Y_i | X_i] = X_i' \beta_0 \Phi\left(\frac{X_i' \beta_0}{\sigma}\right) + \sigma \phi\left(\frac{X_i' \beta_0}{\sigma}\right)$$

ML inference is based on the likelihood function

$$L(\beta, \sigma) = \prod_{i=1}^n \left[1 - \Phi\left(\frac{X_i' \beta_0}{\sigma}\right) \right]^{1(Y_i=0)} \left[\frac{1}{\sigma} \phi\left(\frac{y_i - X_i' \beta}{\sigma}\right) \right]^{1(Y_i>0)} \quad (39)$$

证明. Proof of (39)

Starting with $Y_i = 0$,

$$\begin{aligned} P(Y_i = 0 | X_i) &= P(Y_i^* \leq 0 | X_i) = P(X_i' \beta_0 + u_i^* \leq 0 | X_i) \\ &= P(u_i^* \leq -X_i' \beta_0 | X_i) = P\left(\frac{u_i^*}{\sigma} \leq \frac{-X_i' \beta_0}{\sigma} | X_i\right) = \otimes \\ u_i^* &\sim \mathcal{N}(0, \sigma^2) \Rightarrow \frac{u_i^*}{\sigma} \sim \mathcal{N}(0, 1) \\ \otimes &= \Phi\left(\frac{-X_i' \beta_0}{\sigma}\right) = 1 - \Phi\left(\frac{X_i' \beta_0}{\sigma}\right) \end{aligned}$$

Discrete random variable, we can stop at the cumulative distribution function. Now focusing on $Y_i > 0$

$$\begin{aligned} P(Y_i^* \leq y_i | X_i) &= P(X_i' \beta_0 + u_i^* \leq 0 | X_i) = P(u_i^* \leq y_i - X_i' \beta_0 | X_i) \\ &= P\left(\frac{u_i^*}{\sigma} \leq \frac{y_i - X_i' \beta_0}{\sigma} | X_i\right) = \Phi\left(\frac{y_i - X_i' \beta_0}{\sigma}\right) \end{aligned}$$

For the continuous random variable, we need to find the cumulative density function.

$$\frac{\partial \Phi\left(\frac{y_i - X_i' \beta_0}{\sigma}\right)}{\partial y_i} = \underbrace{\phi\left(\frac{y_i - X_i' \beta_0}{\sigma}\right)}_{\text{density function of } \mathcal{N}(0,1)} \frac{1}{\sigma}$$

□

By **reparameterizing** $L(\theta)$ with $\gamma = \frac{\beta}{\sigma}$ and $\omega = \frac{1}{\sigma}$, the Hessian is negative definite and estimation is easy, but in some situations the estimates may not exist. **Consistency depends on the distributional assumption** but independence is not required and appropriate specification tests are available.

4.3 Correcting OLS

There is a simpler estimator based on OLS. Using the sample with $Y_i > 0$, from truncated data we know that

$$E[Y_i | X_i, Y_i > 0] = X_i' \beta_0 + \sigma \frac{\phi\left(\frac{-X_i' \beta_0}{\sigma}\right)}{1 - \Phi\left(\frac{-X_i' \beta_0}{\sigma}\right)} = X_i' \beta_0 + \sigma \underbrace{\frac{\phi\left(\frac{X_i' \beta_0}{\sigma}\right)}{\Phi\left(\frac{X_i' \beta_0}{\sigma}\right)}}_{\lambda(v)}$$

where $\lambda(v)$ is known as the **inverse Mills ratio**. Now define a dummy variable D_i as

$$D_i = \begin{cases} 1 & \text{if } Y_i > 0 \\ 0 & \text{if } Y_i = 0 \end{cases}$$

Then,

$$P[D_i = 1 | X_i] = P[Y_i > 0 | X_i] = P[Y_i^* > 0 | X_i] = P[u_i^* > -X_i' \beta_0 | X_i] = \Phi\left(\frac{X_i' \beta_0}{\sigma}\right) \quad (40)$$

证明. Proof of (40)

Using the following properties:

- $\Phi(-z) = 1 - \Phi(z)$
- $\phi(-z) = \phi(z)$

$$\begin{aligned} P(Y_i^* > 0 | X_i) &= P(X_i' \beta_0 + u_i^* > 0 | X_i) = P(u_i^* > -X_i' \beta_0 | X_i) = 1 - P(u_i^* \leq -X_i' \beta_0 | X_i) \\ &= 1 - P\left(\frac{u_i^*}{\sigma} \leq \frac{-X_i' \beta_0}{\sigma} | X_i\right) = 1 - \Phi\left(\frac{-X_i' \beta_0}{\sigma}\right) \\ &= 1 - \left[1 - \Phi\left(\frac{X_i' \beta_0}{\sigma}\right)\right] = \Phi\left(\frac{X_i' \beta_0}{\sigma}\right) \end{aligned}$$

□

This motivates the **2-step estimator**.

1. Estimate $\frac{\beta_0}{\sigma}$ by using a **Probit model** for $P[D_i = 1|X_i]$ on the **full sample**. Construct consistent estimates of the **Mills ratio** $\hat{\lambda}_i$ for each observation.
2. **OLS regression** of Y_i on X_i and $\hat{\lambda}_i$ using observations for which $Y_i > 0$. This gives **consistent** estimates for β_0 and σ .

4.4 Sample selection

In many cases, the sample available depends on individual decisions. **Not random $\rightarrow$ subpopulation $\rightarrow$ bias introduced**. Heckman (1979) considered the following model. Suppose we are interested in the following model

$$Y_i = X_i' \beta_0 + \epsilon_i \quad E[\epsilon_i | X_i] = 0$$

but Y_i is observable only when $Z_i = 1$ where

$$Z_i = 1(X_i' \gamma_0 + u_i > 0)$$

If ϵ_i and u_i are **not independent**, the OLS estimation based on the observation with $Z_i = 1$ of β_0 is **inconsistent** because $E[\epsilon_i | X_i, u_i > -X_i' \gamma_0] = E[\epsilon_i | X_i, Z_i = 1] \neq 0$.

Heckman (1976) popularized an estimator of β_0 which is **consistent** under the assumption

$$\begin{bmatrix} u_i \\ \epsilon_i \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho\sigma \\ \rho\sigma & \sigma^2 \end{bmatrix} \right)$$

Assume bivariate normal distribution for u_i and $\epsilon_i \rightarrow$ now we have a closed form distribution. Under this assumption, β_0, γ_0, ρ and σ can be estimated by **maximum likelihood**.

Alternatively, Heckman (1976) used a **two-step estimator Heckit** based on the result

$$\begin{aligned} E[\epsilon_i | X_i, u_i > -X_i' \gamma_0] &= \rho\sigma\lambda(v) \\ E[Y_i | X_i, Z_i = 1] &= X_i' \beta_0 + \rho\sigma\lambda(v) \end{aligned}$$

where $\lambda(v) = \frac{\phi(X_i' \gamma_0)}{\Phi(X_i' \gamma_0)}$. Therefore,

1. $X_i' \gamma_0$ can be estimated by a **probit**. Use and construct consistent estimates of λ_i denoted by $\hat{\lambda}_i$ for each observation
2. β and $\rho\sigma$ are estimated in the **OLS** regression of Y_i on X_i and $\hat{\lambda}_i$.

A **standard t-test** for the significance of $\rho\sigma$ is a valid test for **correlation between u_i and ϵ_i** . In the case where they are correlated, the **covariance matrix** of the second step need to be corrected to account for the estimated regressor. Identification is easier if there are exclusion restrictions. **Can drop some regressors in the second step**.

第 5 章 Univariate Time Series Modelling

5.1 Stochastic processes

定义 5.1.1. A *stochastic process* $\{X_t\}_{t=-\infty}^{+\infty}$ is a sequence of random variables ordered by time.

For each value of t , at each point in time, x_t is drawn from a distribution of X_t 's. Note that the probability density function of $X_t \rightarrow f_{X_t}(x_t)$ depends on t so that each element in the realization may be drawn from a different distribution.

$$E[X_t] = \int_{-\infty}^{+\infty} x_t f_{X_t}(x_t) dx_t = \mu_t$$

$$\text{Var}(X_t) = E[(X_t - \mu_t)^2] = \int_{-\infty}^{+\infty} (x_t - \mu_t)^2 f_{X_t}(x_t) dx_t = \gamma_{0t}$$

$$\text{Cov}(X_t, X_{t-j}) = E[(X_t - \mu_t)(X_{t-j} - \mu_{t-j})] = \gamma_{jt} \rightarrow \text{autocovariance}$$

To make inference, we need to impose some structure.

定义 5.1.2. A stochastic process is **strictly stationary** if for every collection of time indices $1 \leq t_1 < \dots < t_m$ the joint distribution of $(X_{t_1}, \dots, X_{t_m})$ is the same as that of $(X_{t_1+h}, \dots, X_{t_m+h})$ for $h \geq 1$.

定义 5.1.3. A stochastic process is **weakly or covariance stationary** if

- $E[X_t] = \mu$ does not vary with t
- $\text{Var}(X_t) = \gamma_0$ is constant
- for any $j \geq 1$, $\text{cov}(X_t, X_{t-j}) = \text{cov}(X_t, X_{t+j}) = \gamma_j$ depends only on j and not on t

- Strong stationary does not imply weak stationary
- Strong stationary + $E[X_t^2]$ finite implies weak stationary
- Multivariate normal + weak stationary implies strong stationary
- j -th lag autocovariance is denoted as γ_j and j -th lag autocorrelation is denoted as $\rho_j = \frac{\gamma_j}{\gamma_0}$

定理 5.1.4. Wold's Decomposition Theorem: Any covariance stationary process with mean zero can be represented as

$$X_t = \sum_{j=0}^{\infty} \theta_j \epsilon_{t-j} + v_t$$

where $\epsilon_t \sim WN(0, \sigma_\epsilon^2)$, $\theta_0 = 1$ and $\sum_{j=0}^{\infty} \theta_j^2 < \infty$, $E[v_t \epsilon_{t-j}] = 0 \forall j$, and there exists constants $\alpha_0, \alpha_1, \dots$ such that $\text{Var}(\sum_{j=0}^{\infty} \alpha_j v_{t-j}) = 0$.

v_t is the **deterministic** component of X_t . It means that it can be predicted arbitrarily well from a linear function of past values of v_t . The term $\sum_{j=0}^{\infty} \theta_j \epsilon_{t-j}$ is the **indeterministic** component of X_t which is what we want to estimate.

5.2 ARMA processes

Let $a(L)$ be a finite order polynomial in L , $a(L) = 1 - \sum_{i=1}^p a_i L^i$.

$$a(L)^{-1} a(L) = 1 \Leftrightarrow a(L)^{-1} a(L) X_t = X_t$$

$\Downarrow$

$$a(L)^{-1} = \sum_{i=1}^{\infty} b_i L^i$$

Suppose $\Downarrow a(L) = 1 - \rho L$

$$a(L)^{-1} = \sum_{i=0}^{\infty} \rho^i L^i$$

The coefficients of this infinite-order polynomial are **absolutely summable** if $\sum_{i=0}^{\infty} |b_i| < \infty \rightarrow$ necessary condition for a times series to be **stationary**. The necessary and sufficient condition for an inverse to meet the **absolutely summability condition** the characteristic roots of $a(z)$ lie **outside the unit circle**. Find zeroes of the function $a(z) \rightarrow z^*$, to be outside the unit circle we must have $|z^*| > 1$. If $z = a - bi \rightarrow |z| = \sqrt{a^2 + b^2}$.

5.2.1 MA(q)

$$X_t = \sum_{j=0}^q \theta_j \epsilon_{t-j}, \epsilon_t \sim WN(0, \sigma_\epsilon^2), \theta_0 = 1 \Leftrightarrow X_t = \Theta(L) \epsilon_t, \Theta(L) = \sum_{j=0}^q \theta_j L^j$$

By **Wald decomposition theorem** the true model is a $MA(\infty)$. The goal is to truncate the true model to a finite order q .

5.2.2 AR(p)

$$X_t = \sum_{j=1}^p \phi_j X_{t-j} + \epsilon_t, \epsilon_t \sim WN(0, \sigma_\epsilon^2) \Leftrightarrow \Phi(L) X_t = \epsilon_t, \Phi(L) = 1 - \sum_{j=1}^p \phi_j L^j$$

$AR(p)$ is **stationary** if the roots of $\Phi(z)$ are **outside the unit circle**. This model corresponds to a $MA(\infty)$ process.

证明.

$$\Phi(L)X_t = \epsilon_t \Leftrightarrow \Phi(L)^{-1}\Phi(L)X_t = \Phi(L)^{-1}\epsilon_t \Leftrightarrow X_t = (1 + \phi_1 L + \phi_1^2 L^2 + \dots)\epsilon_t$$

□

5.2.3 ARMA(p,q)

$$X_t = \sum_{j=1}^p \phi_j X_{t-j} + \sum_{j=1}^q \theta_j \epsilon_{t-j}, \quad \epsilon_t \sim WN(0, \sigma_\epsilon^2) \Leftrightarrow \Phi(L)X_t = \Theta(L)\epsilon_t$$

$ARMA(p, q)$ is **stationary** if the roots of $\Phi(z)$ are **outside the unit circle**. This model also corresponds to a $MA(\infty)$ process.

证明. Using **method of undetermined coefficients**

$$X_t = \phi_1 X_{t-1} + \epsilon_t + \theta_1 \epsilon_{t-1}$$

$\Downarrow$ stationary

$$X_t = \sum_{i=0}^{\infty} \alpha_i \epsilon_{t-i} \Rightarrow X_{t-1} = \sum_{i=0}^{\infty} \alpha_i \epsilon_{t-i-1}$$

$\Downarrow$ replacing back to ARMA(1,1) model

$$\sum_{i=0}^{\infty} \alpha_i \epsilon_{t-i} = \phi_1 \left(\sum_{i=0}^{\infty} \alpha_i \epsilon_{t-i-1} \right) + \epsilon_t + \theta_1 \epsilon_{t-1}$$

□

$ARMA(p, q)$ is **invertible** if the roots of $\Theta(z)$ are **outside the unit circle**. This model corresponds to a $AR(\infty)$ process.

证明.

$$\Theta(L)^{-1}\Phi(L)X_t = \epsilon_t \Leftrightarrow \left(1 - \sum_{j=1}^{\infty} \alpha_j L^j\right)X_t = \epsilon_t \Leftrightarrow X_t - \sum_{j=1}^{\infty} \alpha_j L^j X_t = \epsilon_t$$

$$\Leftrightarrow X_t - \sum_{j=1}^{\infty} \alpha_j X_{t-j} = \epsilon_t \Leftrightarrow X_t = \sum_{j=1}^{\infty} \alpha_j X_{t-j} + \epsilon_t \rightarrow AR(\infty)$$

□

If $\theta_1 = -\phi_1 \Rightarrow \rho_l = 0$, ARMA(1,1) **reduces to white noise**. Same occurs in ARMA(p,q) processes, **if z^* is a zero of $\Phi(z)$ and $-z^*$ is a zero of $\Theta(z)$** . To avoid **identification problem**, **reduce the model** to $ARMA(p-1, q-1)$. **Choose simpler models**.

5.2.4 Charaterizations

定义 5.2.1. The **autocorrelation function (ACF)** is defined as $\rho_j = \frac{\gamma_j}{\gamma_0}$, $j = 1, 2, \dots$

定义 5.2.2. The **partial autocorrelation** between X_t and X_{t-j} is defined as the coefficient of the variable X_{t-j} in the linear projection of X_t on $X_{t-1}, \dots, X_{t-j}$. Informally, it is defined as a measure of the association between X_t and X_{t-j} whilst taking away the effects of the variables $X_{t-1}, \dots, X_{t-j+1}$ on this relationship for $j \geq 1$.

The linear projection of X_t on $X_{t-1}, \dots, X_{t-j}$ is given by

$$P(X_t | X_{t-1}, \dots, X_{t-j}) = \alpha_j^* + \beta_{j,1}^* X_{t-1} + \dots + \beta_{j,j}^* X_{t-j}$$

$$\text{minimizing } E[(X_t - \alpha_j^* - \beta_{j,1}^* X_{t-1} - \dots - \beta_{j,j}^* X_{t-j})^2]$$

So the partial autocorrelation of order j is $\beta_{j,j}^*$. The goal is to minimize the mean of the square of the difference.

The **Euler Walker equations** give us the easy way to calculate PACF for any stochastic process

$$\beta_{1,1}^* = \rho_1$$

$$\beta_{2,2}^* = \frac{(\rho_2 - \rho_1^2)}{(1 - \rho_1^2)}$$

$$\beta_{j,j}^* = \frac{\rho_j - \sum_{i=1}^{j-1} \beta_{j-1,i}^* \rho_{j-i}}{1 - \sum_{i=1}^{j-1} \beta_{j-1,i}^* \rho_i}, \quad j = 3, 4, \dots$$

$$\beta_{j,i}^* = \beta_{j-1,i}^* - \beta_{j,j}^* \beta_{j-i,j-i}^*, \quad i = 1, 2, \dots, j-1$$

- For **MA(q) processes**, **ACF significant** up to its lag, PACF shows direct decay if $\theta < 0$ or exponential decay if $\theta > 0$.
- For **AR(p) processes**, ACF shows direct decay if $\phi > 0$ and exponential decay if $\phi < 0$, **PACF significant** up to its lag.
- For **ARMA(p,q) processes**, both **ACF and PACF exponentially decay**.

5.2.5 ARIMA(p,d,q)

定义 5.2.3. If a stochastic process X_t must be differentiated exactly d times to admit auto-regressive and moving-average representations that are both absolutely summable, then the series is $I(d)$ and we write $X_t \sim I(d)$. The process X_t is said to be **integrated** of order d .

If $\Delta^d X_t$ follows an $ARMA(p, q)$ model

$$\Phi(L)\Delta^d X_t = x + \Theta(L)\epsilon_t$$

with all roots of $\Phi(L)$ and $\Theta(L)$ outside the unit circle, then X_t follows an autoregressive integrated moving average model of order (p,d,q) . The $ARIMA(p,d,q)$ is non-stationary $ARMA(p+d,q)$ where the autoregressive polynomial $\Phi^*(L) = \Phi(L)(1-L)^d$ has d unit roots. Therefore testing procedures to **determine d** focus on the **number of autoregressive unit roots**.

- If $X_t \sim I(0)$, shock ϵ_t has a transient decaying effect on X_{t+k} as $k \rightarrow \infty$. X_t fluctuates around the mean and **ACF has a cut-off point** or **decays exponentially**.
- If $X_t \sim I(1)$, shock ϵ_t has a permanent effect on X_{t+k} as $k \rightarrow \infty$. X_t is not mean-reversion and displays a time-varying trend. **ACF is not defined** but **sample ACF decays slowly**, approximately linear.

5.2.6 Box Jenkins Methodology

1. Transform the data so that the assumption of covariance stationarity is reasonable
2. Make an initial guess of the propose $ARMA(p,q)$ model
3. Estiamte the parameters of the proposed model
4. Perform diagnostic analysis to confirm that the proposed model adequately describes the data

5.2.6.1 Identification of model : sample autocorrelation

$$\hat{\gamma}_j = \frac{1}{T} \sum_{t=j+1}^T (X_t - \bar{X})(X_{t-j} - \bar{X}), \quad j \geq 0 \Rightarrow \hat{\rho}_j = \frac{\hat{\gamma}_j}{\hat{\gamma}_0}, \quad j = 1, 2, \dots$$

If X_t is i.i.d. for all t then $\rho_j = 0$ and we have $\sqrt{T}\hat{\rho}_j \xrightarrow{d} N(0,1) \quad \forall j = 1, \dots, k$

$$\begin{aligned} H_0 : \rho_j &= 0 & H_1 : \rho_j &\neq 0 \\ \left| \sqrt{T}\hat{\rho}_j \right| &> z_{\frac{\alpha}{2}} & \rightarrow \text{reject } H_0 \end{aligned}$$

We can also use the **Box-Pierce Portmonteau statistic** where $Q_k = T \sum_{j=1}^k \hat{\rho}_j^2 \xrightarrow{d} \chi^2(k)$

$$\begin{aligned} H_0 : \rho_j &= 0, \quad j = 1, \dots, k & H_1 : \text{there is at least one } \rho_j &\neq 0 \\ Q_k &> c_\alpha & \rightarrow \text{reject } H_0 \end{aligned}$$

Another option is the **Ljung and Box** where $Q_k^* = T(T+2) \sum_{j=1}^k \frac{\hat{\rho}_j^2}{T-j} \xrightarrow{d} \chi^2(k)$

$$\begin{aligned} H_0 : \rho_j &= 0, \quad j = 1, \dots, k & H_1 : \text{there is at least one } \rho_j &\neq 0 \\ Q_k^* &> c_\alpha & \rightarrow \text{reject } H_0 \end{aligned}$$

5.2.6.2 Identification of model : sample partial autocorrelation

If X_t is i.i.d. for all t then $\beta_{jj} = 0$ and we have $\sqrt{T}\hat{\beta}_{jj} \xrightarrow{d} N(0, 1) \forall j = 1, \dots, k$

$$H_0 : \beta_{jj} = 0 \quad H_1 : \beta_{jj} \neq 0$$

$$\left| \sqrt{T}\hat{\beta}_{jj} \right| > z_{\frac{\alpha}{2}} \rightarrow \text{reject } H_0$$

5.2.6.3 Maximum Likelihood Estimation

In time series, $f(x_1, \dots, x_T; \gamma) \neq \prod_{t=1}^T f(x_t; \gamma)$ because the random variables in sample $(X_1, \dots, X_T)$ are **not iid**. One possible solution is the **conditional factorization of the density function**.

$$f(x_T, \dots, x_1; \gamma) = \left(\prod_{t=2}^T f(x_t | F_{t-1}; \gamma) \right) f(x_1; \gamma)$$

where $F_t = (x_t, \dots, x_1)$ denotes information available at time t . We then have

$$\log \mathcal{L}(\gamma | X_1, \dots, X_T) = \left(\sum_{t=2}^T \log f(x_t | F_{t-1}; \gamma) \right) + \log f(x_1; \gamma) \rightarrow \text{exact log-likelihood function}$$

$$\log \mathcal{L}^*(\gamma | X_1, \dots, X_T) = \sum_{t=2}^T \log f(x_t | F_{t-1}; \gamma) \rightarrow \text{conditional log-likelihood function}$$

The assumption of **normality** is imposed.

- We get the **exact likelihood estimator** (EML) if we **maximize the exact log-likelihood function**. We get the **conditional maximum likelihood estimator** (CML) $\hat{\gamma}_{CML}$ if we **maximize the conditional log-likelihood function**.
- Both estimators are **consistent and asymptotically normal** under regularity conditions

$$\sqrt{T}(\hat{\gamma}_{CML} - \gamma_0) \xrightarrow{d} N(0, A_0^{-1}) \quad A_0 = E\left[-\frac{\partial^2 l_t(\gamma_0)}{\partial \gamma \partial \gamma'}\right]$$

but they will not yield the same estimates in finite samples.

5.2.6.4 Diagnostic testing

To check if **residuals $\hat{\epsilon}_t$ have white noise properties**.

- Check the **sample ACF and sample PACF** of the residuals
- Use **Box-Pierce and Ljung-Box** statistics applied to the residuals

Test for **serial correlation** using the **Breusch Godfrey** Lagrange multiplier statistic. To test for white noise against r -th order autocorrelation in the residuals in a $AR(p)$ model amounts to test $\beta_1 = \dots = \beta_r = 0$ in auxiliary regression

$$\hat{\epsilon}_t = \alpha_0 + \alpha_1 X_{t-1} + \dots + \alpha_p X_{t-p} + \beta_1 \hat{\epsilon}_{t-1} + \dots + \beta_r \hat{\epsilon}_{t-r} + e_t$$

$$LM = T \times R^2 \xrightarrow{d} \chi^2(r).$$

The less arbitrary way to test for the **goodness of fit** of the model is to use the **information criterion** such as the Akaike information criterion (AIC) or the Bayesian (Swirartz) information criterion (BIC). Given several models, we should choose the one with the **lowest information criteria**.

5.3 Forecasting

$F_s = \{X_s, X_{s-1}, \dots\}$ information on process X_t up to s . The **conditional expectation** $E[X_t|F_s]$, $s < t$, is best under squared error loss predictor of X_t given F_s is

$$E[(X_t - E[X_t|F_s])^2] \leq E[(X_t - g(F_s))^2]$$

for all functions $g(F_s)$. The $g(\cdot)$ function might not be linear. The **best linear predictor** $P(X_t|F_s)$, $s < t$ is

$$E[(X_t - P(X_t|F_s))^2] \leq E[(X_t - g(F_s))^2]$$

for all linear functions $g(F_s)$.

定义 5.3.1. The process X_t is **martingale** if $E[X_{t+1}|F_t] = X_t$ for all t .

定义 5.3.2. The process X_t is a **martingale difference sequence** if $E[Y_{t+1}|F_t] = 0$ for all t .

If X_t is a martingale, then $Z_t = X_t - X_{t-1}$ is a martingale difference sequence.

If we assume **errors are iid** or a martingale difference sequence, $E[\epsilon_{t+1}|F_t] = 0 \forall t$, we are able to estimate $E[X_t|F_s]$. And we are able to show that the estimator for best forecast is given by

$$E[X_{t+l}] = c + \sum_{j=1}^p \phi_j E[X_{t+l-j}] + \sum_{j=1}^q \theta_j \epsilon_t(l-j)$$

$$E[X_{t+l-j}] = X_{t+l-j}, j \geq l \quad \epsilon_t(l-j) = \begin{cases} \epsilon_{t+l-j} & j \geq l \\ 0 & j < l \end{cases}$$

The **variance of the prediction errors** $e_t(l) = X_{t+l} - E[X_{t+l}]$ is obtained from the $MA(\infty)$ representation of the $ARMA(p, q)$ process $X_t = E[X_t] - \sum_{j=0}^{\infty} \phi_j \epsilon_{t-j}$. Assuming $\text{Var}(\epsilon_t) = \sigma_\epsilon^2$ for all t ,

$$\text{Var}(e_t(l)) = \sigma_\epsilon^2 \sum_{j=0}^{l-1} \phi_j^2$$

for stationary process as $l \rightarrow \infty$ we have

$$E[X_{t+l}] \rightarrow E[X_t] \quad \text{Var}(e_t(l)) \rightarrow \text{Var}(X_t)$$

For $I(1)$ processes, the above methods are applied to ΔX_t , yielding $E[\Delta X_{t+l}|F_t]$. The forecasts are given by

$$E_t[X_{t+l}] = X_t + \sum_{i=1}^l E_t[\Delta X_{t+i}]$$

To **compare the forecasts** among different ARMA/ARIMA models

1. leave the last observations of the time series out of the estimation of the models
2. produce forecasts for these observations for each model
3. choose the model that yields the **minimum value of the sample mean square prediction error** among the estimated models

第 6 章 Multivariate Time Series Models

Multivariate time series models allow for investigation of dynamic relationships between a set of variables without imposing endogeneity or exogeneity restrictions. The generalization of the **Autoregressive Moving Average Model** to the multivariate case is the **Vector Autoregressive Moving Average Model** (VARMA). A special case of the VARMA is the **Vector Autoregressive Model** (VAR). It is used to capture the evolution and the interdependencies between multiple time series. All the variables are treated symmetrically by including for each variable an equation explaining its evolution based on its own lags and the lags of all other variables in the model.

6.1 Stationary multivariate time series

Let $X_t = (X_{1,t}, \dots, X_{k,t})'$ be a k dimensional vector time series. We can define the **mean** as

$$\mu_t = \begin{bmatrix} E[X_{1,t}] \\ \vdots \\ E[X_{k,t}] \end{bmatrix}$$

the **variance matrix** as

$$\Gamma_{t,t} = \begin{bmatrix} \text{Var}(X_{1,t}) & \text{COV}(X_{1,t}, X_{2,t}) & \cdots & \text{COV}(X_{1,t}, X_{k,t}) \\ \text{COV}(X_{2,t}, X_{1,t}) & \text{Var}(X_{2,t}) & \cdots & \text{COV}(X_{2,t}, X_{k,t}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{COV}(X_{k,t}, X_{1,t}) & \text{COV}(X_{k,t}, X_{2,t}) & \cdots & \text{Var}(X_{k,t}) \end{bmatrix}$$

and the **autocovariance matrix** as

$$\Gamma_{t,t-l} = \begin{bmatrix} \text{COV}(X_{1,t}, X_{1,t-l}) & \text{COV}(X_{1,t}, X_{2,t-l}) & \cdots & \text{COV}(X_{1,t}, X_{k,t-l}) \\ \text{COV}(X_{2,t}, X_{1,t-l}) & \text{COV}(X_{2,t}, X_{2,t-l}) & \cdots & \text{COV}(X_{2,t}, X_{k,t-l}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{COV}(X_{k,t}, X_{1,t-l}) & \text{COV}(X_{k,t}, X_{2,t-l}) & \cdots & \text{COV}(X_{k,t}, X_{k,t-l}) \end{bmatrix}$$

note this is not symmetric.

定义 6.1.1. X_t is **weakly stationary** if for all t and l

- $\mu_t = \mu$
- $\Gamma_{t,t-l} = \Gamma_l = \Gamma'_{-l}$

We define the **cross correlation** as

$$\rho_{ij}(l) = \text{Corr}[X_{it}, X_{j,t-l}] = \frac{\Gamma_{ij}(l)}{\sqrt{\Gamma_{ii}(0)\Gamma_{jj}(0)}}$$

$\Downarrow$ collecting cross-correlations from matrix

$$\rho_l = D^{-1}\Gamma_l D^{-1} \quad D = \text{diag}\{\sqrt{\Gamma_{11}(0)}, \dots, \sqrt{\Gamma_{kk}(0)}\}$$

Stationarity implies $\rho_l = \rho'_{-l}$. The **ACF** of X_{kt} are the diagonal elements of ρ_l .

定义 6.1.2. Multivariate White Noise ϵ_t is a stationary process with

- $E[\epsilon_t] = \vec{0}$
- $\text{Var}(\epsilon_t) = E[\epsilon_t \epsilon_t'] = \Omega \rightarrow$ a constant variance-covariance matrix
- $\text{Cov}(\epsilon_t, \epsilon_s) = E[\epsilon_t \epsilon_s'] = 0$ for $s \neq t \rightarrow$ uncorrelated

Under the assumption of **multivariate iid** thus $\rho_l = 0, \forall l \neq 0$ we have $\sqrt{T}\hat{\rho}_{ij}(l) \xrightarrow{d} N(0, 1)$

$$H_0 : X_t \text{ is iid} \quad H_1 : X_t \text{ not iid}$$

$$Q_k(m) \xrightarrow{d} \chi^2(k^2 m)$$

定理 6.1.3. Multivariate Wold Decomposition Theorem: If the k -variate X_t time series process is weakly stationary, then it has the representation

$$X_t = \sum_{s=0}^{\infty} \Lambda_s \epsilon_{t-s} + W_t$$

where the $k \times k$ matrices Λ_s are such that $\Lambda_0 = I_k$, $\sum_{s=0}^{\infty} \Lambda_s \Lambda_s'$ converges, the process ϵ_t is a k variate white noise process and $W_t \in \mathbb{R}^k$ is a linear deterministic process, that there exists a k vector x_0 and $k \times k$ matrices C_s such that $W_t = c_0 + \sum_{s=0}^{\infty} C_s W_{t-s}$ and $E[\epsilon_t W_{t-m}] = 0$ for $m = 0, \pm 1, \pm 2, \dots$

Similar to the univariate case, **inversion** of polynomials in L means

$$H(L) = I - \sum_{i=1}^p H_i L^i \Leftrightarrow H(L)^{-1} H(L) = I \Rightarrow H(L)^{-1} = \sum_{i=0}^{\infty} B_i L^i$$

$$\text{suppose } H(L) = I - \Pi L \Rightarrow H(L)^{-1} = \sum_{i=0}^{\infty} \Pi^i L^i$$

The coefficients of the infinite order polynomial $H(L)^{-1} = \sum_{i=0}^{\infty} B_i L^i$ are **absolutely summable** if $\sum_{i=0}^{\infty} |b_{kli}| < \infty$.

The necessary and sufficient condition for **stationarity** in the multivariate case

- If the roots of the characteristic equation

$$\left| I\lambda^p - \sum_{l=1}^p H_l \lambda^{p-l} \right| = 0$$

are **inside the unit circle**.

- equivalently if all values of z satisfying

$$\left| I - \sum_{l=1}^p H_l z^l \right| = 0$$

are **outside the unit circle**.

6.2 Vector Autoregressive models

Any k the Vector autoregressive model of order p , **VAR(p)**, is defined as

$$X_t = \phi_0 + \sum_{l=1}^p \Phi_l X_{t-l} + \epsilon_t$$

which can also be written as a system of equations

$$\begin{aligned} X_{1,t} &= \phi_{10} + \sum_{l=1}^p \sum_{j=1}^k \Phi_{1j}(l) X_{j,t-l} + \epsilon_{1t} \\ X_{2,t} &= \phi_{20} + \sum_{l=1}^p \sum_{j=1}^k \Phi_{2j}(l) X_{j,t-l} + \epsilon_{2t} \\ &\vdots \\ X_{k,t} &= \phi_{k0} + \sum_{l=1}^p \sum_{j=1}^k \Phi_{kj}(l) X_{j,t-l} + \epsilon_{kt} \end{aligned}$$

Notice that

$$\begin{aligned} X_t &= \phi_0 + \left(\sum_{l=1}^p \Phi_l L^l \right) X_t + \epsilon_t \\ (I_k - \sum_{l=1}^p \Phi_l L^l) X_t &= \phi_0 + \epsilon_t \\ \Phi(L) X_t &= \phi_0 + \epsilon_t \\ \Downarrow \text{stationary} \\ X_t &= \Phi(L)^{-1} (\phi_0 + \epsilon_t) \\ &= \left(\sum_{j=0}^{\infty} \Psi_j L^j \right) (\phi_0 + \epsilon_t) \\ &= \sum_{j=0}^{\infty} \Psi_j \phi_0 + \sum_{j=0}^{\infty} \Psi_j \epsilon_{t-j} \rightarrow \text{Vector Moving Average} \end{aligned}$$

6.2.1 Impulse response functions

Consider the $MA(\infty)$ representation of the $VAR(p)$ process

$$X_t = c + \sum_{r=0}^{\infty} \Psi_r \epsilon_{t-r} \quad c = \sum_{r=0}^{\infty} \Psi_r \phi_0$$

Notice that

$$\frac{\partial X_{t+l}}{\partial \epsilon'_t} = \Psi_l \Rightarrow \frac{\partial X_{i,t+l}}{\partial \epsilon_{j,t}} = \Psi_{i,j}(l)$$

定义 6.2.1. The *impulse response function* describes the response of $X_{i,t+l}$ to a one-time unit change in $\epsilon_{j,t}$.

Usually we **multiply $\Psi_{i,j}(l)$ by the standard deviation of $\epsilon_{j,t}$** so we obtain the response of $X_{i,t+l}$ to a one-time change in $\epsilon_{j,t}$ of $\text{Var}(\epsilon_{j,t})^{1/2}$ unit.

6.2.2 Estimation

Assume

$$X_t = \phi_0 + \sum_{l=1}^p \Phi_l X_{t-l} + \epsilon_t, \quad t = 1, \dots, T \quad \epsilon_t \sim iid N(0, \Omega)$$

We shall **condition on the p first observations** and **derive the conditional likelihood function** for $X_1, \dots, X_T$.

Rewriting the equation

$$\begin{aligned} X_t &= B' Z_t + \epsilon_t, \quad t = p+1, \dots, T \\ Z_t &= (1, X'_{t-1}, \dots, X'_{t-p})' \quad B' = [\phi_0, \Phi_1, \dots, \Phi_p] \quad Z_t^* = (X'_t, \dots, X'_{t-p})', \quad t \geq p+1 \\ &\Downarrow \\ X_t | Z_t^* &\sim N(B' Z_t, \Omega) \\ &\Downarrow \text{conditional MLE} \\ \hat{B} &= \left[\sum_{t=p+1}^T x_t z'_t \right] \left[\sum_{t=p+1}^T z_t z'_t \right]^{-1} \\ \hat{\Omega} &= \frac{1}{T} \sum_{t=p+1}^T \hat{\epsilon}_t \hat{\epsilon}'_t \quad \hat{\epsilon}_t = x_t - \hat{B}' z_t \end{aligned}$$

The conditional MLE of $\hat{B}$ is obtained by **applying OLS separately to each equation**.

We can show that $\sqrt{T}(\hat{b} - b) \xrightarrow{d} N(0, \Sigma \otimes E[z_t z'_t])$. Thus the **likelihood ratio** to test restrictions is

$$\begin{aligned} H_0 : r(\theta) &= 0 \quad H_1 : r(\theta) \neq 0 \\ LR &= T(\log |\hat{\Omega}_r| - \log |\hat{\Omega}|) \xrightarrow{d} \chi^2(h) \quad h = \dim[r(\theta)] \end{aligned}$$

6.2.3 Bivariate Granger causality

A scalar variable X **Granger causes** another scalar variable Y if Y can be **better predicted** using the histories of both X and Y than it can using the history of Y alone.

定义 6.2.2. X *fails to Granger cause* Y if

$$MSE[\hat{E}[Y_{t+s}|Y_t, Y_{t-1}, \dots]] = MSE[\hat{E}[Y_{t+s}|X_t, X_{t-1}, \dots, Y_t, Y_{t-1}, \dots]]$$

where MSE is the mean square error of prediction: $MSE(\hat{E}(\cdot)) = E[(Y_{t+s} - \hat{E}[\cdot])^2]$

Let

$$y_t = c + \sum_{i=1}^p [\alpha_i x_{t-i} + \beta_i y_{t-i}] + u_t$$

under the assumptions that

- regressors are contemporaneously exogenous
- regressors are contemporaneously homoskedastic
- no autocorrelation between the error terms

x **fails to Granger cause** y if

$$H_0 : \alpha_i = 0, i = 1, \dots, p$$

$$S = \frac{T(RRS_0 - RRS_1)}{RRS_1} \xrightarrow{d} \chi^2(p)$$

$$\begin{cases} RRS_0 \rightarrow y_t = c + \sum_{i=1}^p \beta_i y_{t-i} + u_t \\ RRS_1 \rightarrow y_t = c + \sum_{i=1}^p [\alpha_i x_{t-i} + \beta_i y_{t-i}] + u_t \end{cases}$$

$$S > c_\alpha \rightarrow \text{Reject } H_0 \rightarrow x \text{ Granger causes } y$$

6.2.4 Specification tests

- For **misspecification** the **Multivariate Q-statistic** can be applied to the residuals with asymptotic $\chi^2(k^2(m-p))$ distribution
- For **serial correlation** we can apply the **vector LM tests**
- To select the **lag length**, we can use the **information criterion**. Choose the one based on minimization.

6.2.5 Special case : Structural VAR

Structural VAR (SVAR) allows **contemporaneous relationships** between elements of X_t .

$$B_0 X_t = c_0 + B_1 X_{t-1} + B_2 X_{t-2} + \cdots + B_p X_{t-p} + U_t$$

$$B_0 X_t = c_0 + \sum_{l=1}^p B_l X_{t-l} + U_t$$

$\Downarrow$ if invertable

$$X_t = \phi_0 + \sum_{l=1}^p \Phi_l X_{t-l} + \epsilon_t \rightarrow \text{reduced form of VAR}$$

$$\phi_0 = B_0^{-1} c_0 \quad \Phi_l = B_0^{-1} B_l \quad \epsilon_t = B_0^{-1} U_t$$

The SVAR has k^2 more parameters than the VAR model so we need k^2 **more restrictions** in order to identify the parameters of the SVAR. **This is why we don't want so many parameters.** Essentially a **necessary condition for identification** requires B_0 and D to have **no more unknown elements than Ω** which is $\frac{k(k+1)}{2}$. This condition is the **order condition** for identification.

- **Normization restrictions**: assign the coefficient of 1 to X_{it} in each equation (k)
- **Covariance matrix restrictions**: specifying D to be diagonal (k)
- **Cholesky Decomposition**: B_0 is lower triangular.

定义 6.2.3. Based on a **Cholesky type decomposition** of a positive definite matrix. Any symmetric positive definite matrix A can be decomposed as $A = LGL'$ where G is a diagonal matrix and L is a lower triangular matrix with 1's in the diagonal.

- Impose restrictions based on economic theory

6.2.6 Forecasting

Begin with a VAR(p) model,

$$\begin{aligned}
 X_t &= \phi_0 + \sum_{i=1}^p \Phi_i X_{t-i} + \epsilon_t \rightarrow \text{stationary} \\
 X_{h+l} &= \phi_0 + \sum_{i=1}^p \Phi_i X_{h+l-i} + \epsilon_{h+l} \\
 E_h[X_{h+l}] &= E_h \left[\phi_0 + \sum_{i=1}^p \Phi_i X_{h+l-i} + \epsilon_{h+l} \right] \\
 &= \phi_0 + \sum_{i=1}^p \Phi_i E_h[X_{h+l-i}] + \underbrace{E_h[\epsilon_{h+l}]}_{=0, \text{ tower property}} \\
 X_h(l) &= \phi_0 + \sum_{i=1}^p \Phi_i X_h(l-i)
 \end{aligned}$$

The **variance** of the forecasting error is given by

$$\text{Var}(e_h(l)) = \sum_{j=0}^{l-1} \Psi_j \Omega \Psi_j'$$

第 7 章 Nonparametric and Semiparametric Estimators

A **parametric estimator** is used to estimate **finite-dimensional parameters** of a model where these parameter values uniquely identify the conditional probability of the dependent variables. A **nonparametric estimator** is used to estimate features of a model **without making any assumptions** on the population distribution. It is often used to **estimate functions**, such as the probability density function or a conditional mean without assuming strong conditions such as normality. These are denoted as **infinite dimensional parameters**. A **semiparametric estimator** is used to estimate **finite-dimensional parameters** but these do not uniquely identify the conditional probability distribution of the dependent variables.

The same estimator can be seen as parametric, nonparametric, and semiparametric depending on the context. Consider the linear regression model

$$y = x'\beta_0 + u$$

and the Ordinary Least Squares Estimator (OLS).

- OLS is **parametric** if we assume $u|x \sim N(0, \sigma^2)$ and obtain the estimator via ML.
- OLS is **semiparametric** if we assume $E[u|x] = 0$ thus $E[y|x] = x'\beta_0$. we do not assume normality of the error term, but assume a specific form for the mean of y conditional on x .
- OLS is **nonparametric** if our objective is to estimate the best linear predictor coefficients. That is, we want to estimate the value of β that minimizes $E[(y - x'\beta)^2]$. *We don't assume anything. Just finding the function to minimize $E[(y - x'\beta)^2]$ but no assumption on the distribution of the data.*

7.1 Nonparametric estimation

Suppose $\{y_i\}_{i=1}^n$ are iid where y_i is a random variable with a continuous distribution. Our objective is to **estimate the density function of $y \rightarrow f(y)$** .

7.1.1 Crude estimator

A **crude estimator** of the density of y is the histogram. The shape of the histogram depends on the **bin width**, on the **number of bins**, and on the chosen **origin**. Fixing these parameters, the density of y at $y = y_0$ given by the histogram is

$$\hat{f}(y_0) = \frac{1}{n} \frac{\text{number of observations in the same bin as } y_0}{\text{width of the bin containing } y_0}$$

If we define h as the bin width and let y^* denote the midpoint of the interval containing y_0 , the histogram estimator is given by

$$\hat{f}(y_0) = \frac{1}{nh} \sum_{j=1}^n 1\left(-\frac{1}{2} \leq \frac{y_j - y^*}{h} < \frac{1}{2}\right)$$

where $1(A) = \begin{cases} 1 & \text{if } A \text{ is true} \\ 0 & \text{if } A \text{ is not true} \end{cases} \rightarrow \text{Bernoulli distribution. Note the estimator is **not continuous at } y_0**$.

7.1.2 Naive density estimator

From the definition of a probability density, we have

$$\begin{aligned} f(y_0) &= \lim_{h \rightarrow 0} \frac{F(y_0 + \frac{h}{2}) - F(y_0 - \frac{h}{2})}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} P(y_0 - \frac{h}{2} < y < y_0 + \frac{h}{2}) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} E \left[1 \left(y_0 - \frac{h}{2} < y < y_0 + \frac{h}{2} \right) \rightarrow \text{Bernoulli} \right] \\ &= 1 \times P \left(y_0 - \frac{h}{2} \leq y < y_0 + \frac{h}{2} \right) + 0 \times \left(1 - P \left(y_0 - \frac{h}{2} < y < y_0 + \frac{h}{2} \right) \right) \\ &= P \left(y_0 - \frac{h}{2} \leq y < y_0 + \frac{h}{2} \right) \end{aligned}$$

Then use the averages to find the expected values. Hence the estimator of $f(y_0)$ given by the **analogy principle** is the **naive density estimator**

$$\begin{aligned} \hat{f}(y_0) &= \frac{1}{nh} \sum_{j=1}^n 1 \left(y_0 - \frac{h}{2} \leq y < y_0 + \frac{h}{2} \right) \\ &= \frac{1}{nh} \sum_{j=1}^n 1 \left(-\frac{1}{2} \leq \frac{y_j - y_0}{h} < \frac{1}{2} \right) \end{aligned}$$

For the estimator to be **consistent and ensures convergence** in iid samples, we need $n \rightarrow \infty$, $nh \rightarrow \infty$ **and** $h \rightarrow 0$.

证明. $h \rightarrow 0$

$$\begin{aligned} E\hat{f}(y_0) &= E \left[\frac{1}{nh} \sum_{j=1}^n 1 \left(y_0 - \frac{h}{2} \leq y_j < y_0 + \frac{h}{2} \right) \right] \\ &= \frac{1}{nh} \sum_{j=1}^n E \left[1 \left(y_0 - \frac{h}{2} \leq y < y_0 + \frac{h}{2} \right) \right] (y_j \rightarrow y \text{ because iid}) \\ &= \frac{n}{nh} E \left[1 \left(y_0 - \frac{h}{2} \leq y < y_0 + \frac{h}{2} \right) \right] \\ &\quad \underbrace{\hspace{10em}}_{f(y_0) \text{ definition}} \\ &= \frac{1}{h} \left[F \left(y_0 + \frac{h}{2} \right) - F \left(y_0 - \frac{h}{2} \right) \right] \end{aligned}$$

So, from $f(y_0)$ definition,

$$\frac{1}{h} \left[F\left(y_0 - \frac{h}{2}\right) - F\left(y_0 + \frac{h}{2}\right) \right] \rightarrow f(y_0) \text{ if } h \rightarrow 0$$

□

证明. $nh \rightarrow \infty$

$$\begin{aligned} \text{Var}(\hat{f}(y_0)) &= \text{Var} \left[\frac{1}{nh} \sum_{j=1}^n 1 \left(y_0 - \frac{h}{2} \leq y_j < y_0 + \frac{h}{2} \right) \right] \\ &= \frac{1}{n^2 h^2} \sum_{j=1}^n \text{Var} \left[1 \left(y_0 - \frac{h}{2} \leq y_j < y_0 + \frac{h}{2} \right) \right] \text{ (variance of sum = sum of variance)} \\ &= \frac{n}{n^2 h^2} \underbrace{\text{Var} \left[1 \left(y_0 - \frac{h}{2} \leq y < y_0 + \frac{h}{2} \right) \right]}_{\text{Var[Bernoulli]} = \theta(1-\theta)} \\ &= \frac{1}{nh^2} P \left(y_0 - \frac{h}{2} \leq y < y_0 + \frac{h}{2} \right) \times \underbrace{\left[1 - P \left(y_0 - \frac{h}{2} \leq y < y_0 + \frac{h}{2} \right) \right]}_{\leq 1} \end{aligned}$$

and because $0 \leq \text{Var}(\hat{f}(y_0))$

$$0 \leq \text{Var}(\hat{f}(y_0)) \leq \frac{1}{nh} \underbrace{\frac{P \left(y_0 - \frac{h}{2} \leq y < y_0 + \frac{h}{2} \right)}{h}}_{f(y_0) \rightarrow \text{constant if } h \rightarrow 0}$$

Thus we conclude $\frac{1}{nh} \rightarrow 0 \Rightarrow nh \rightarrow \infty$

□

Consequently, $\lim \hat{f}(y_0) \xrightarrow{P} f(y_0)$. If $\lim E[\hat{\theta}] = \theta_0$ and $\lim \text{Var}[\hat{\theta}] = 0$ then $\lim \hat{\theta} \xrightarrow{P} \theta_0$. Like the histogram estimator, the naive estimator is **not continuous at y_0** .

7.1.3 Kernel estimator

A continuous and smooth estimator can be obtained by replacing the function $K[z] = 1 \left(-\frac{1}{2} \leq z < \frac{1}{2} \right)$ by another **kernel function**.

定义 7.1.1. A **Kernel function** is any function $K(\cdot)$ such that

$$\int_{\mathbb{R}} K(u) du = 1$$

(Rosenblatt, 1956)

We do not require $K(\cdot) \geq 0$ although this assumption ensures that the estimator is non-negative. Many kernels are available but a popular choice is to use $K[z] = \phi(z)$, the density function of the standard normal distribution. In general, for a bandwidth $h > 0$, the **kernel density function** is given by

$$\hat{f}(y_0) = \frac{1}{nh} \sum_{j=1}^n K \left(\frac{y_i - y_0}{h} \right)$$

The choice of the kernel **affects the bias** $E[\hat{f}(y_0)] - f(y_0)$. **Consistency** requires $n \rightarrow \infty$, $nh \rightarrow \infty$, and $h \rightarrow 0$.

This can be generalized to d -dimensional data using the multivariate kernel function $K_d(\cdot)$ such that $\int_{\mathbb{R}^d} K_d(u) du = 1$. To ensure **consistency**, we need to assume $n \rightarrow \infty$, $h \rightarrow 0$, and $nh^d \rightarrow \infty$. In this case the variance of the estimator goes to zero as $nh^d \rightarrow \infty$. Notice the **curse of dimensionality**, the larger the value of d , the slower the rate of convergence. The norm $\|\hat{\beta} - \beta_0\| \leq \frac{c}{N^{1/2}} \rightarrow$ goes to zero much slower than **parametric**. Under additional conditions it is also possible to ensure **asymptotic normality** but the rate of convergence is slower than rate $n^{-1/2}$ and the estimator is **biased**. The rate of convergence depends on $K_d(\cdot)$ but other authors say this choice is relatively unimportant. The **bandwidth of h** affects the **bias and variance** of the estimator, as h decreases, so does the bias; however, the variance increases.

The choice of **bandwidth** is critical for the result. **Too large** means **over-smoothing** and eliminates important features of the function being estimated. **Too small** means **under-smoothing** and the important features of the function become obscured by noise. Choices of implementation rules include

- Automatic rules
- Graphically analyzing the effects of varying the bandwidth
- Minimizing selection criterion
- Silverman's rule of thumb $\rightarrow$ popular choice for the Gaussian kernel

$$h = \min \left\{ \sigma, \frac{IQR}{1.34} \right\} \frac{0.9}{n^{1/5}}$$

7.1.4 Kernel regression : discrete x_i

The kernel density estimator can be used for **kernel regression**. It is particularly useful for exploratory analysis of the data as it may be convenient not to assume a functional form for the relation between two variables.

Let $\{(y_i, x_i)\}_{i=1}^n$ be iid where y_i and x_i are random variables with continuous distributions and suppose we want to estimate $E[y_i|x_i] = m(x_i)$. We can write this as

$$y_i = m(x_i) + \epsilon_i$$

where $E[\epsilon_i|x_i] = 0$.

证明. $E[\epsilon_i|x_i] = 0 \rightarrow y_i = m(x_i) + \epsilon_i$

$$\begin{aligned}
 \epsilon_i &= y_i - E[y_i|x_i] \\
 E[\epsilon_i|x_i] &= E[y_i - E[y_i|x_i]|x_i] \\
 &= E[y_i|x_i] - E[E[y_i|x_i]|x_i] \\
 &= E[y_i|x_i] - E[y_i|x_i] = 0 \\
 \Rightarrow y_i &= \underbrace{E[y_i|x_i]}_{m_i} + \epsilon_i
 \end{aligned}$$

□

If we are interested in estimating $m(x_j)$ at a particular value of x and if **multiple observations of y** were available for that particular value of x , we could use

$$\begin{aligned}
 \hat{m}(x_0) &= \frac{\sum_{j=1}^n 1(x_j = x_0)y_j}{\sum_{j=1}^n 1(x_j = x_0)} = \sum_{j=1}^n \frac{1(x_j = x_0)}{\sum_{j=1}^n 1(x_j = x_0)} \\
 &= m(x_0) + \frac{\sum_{j=1}^n 1(x_j = x_0)\epsilon_j}{\sum_{j=1}^n 1(x_j = x_0)}
 \end{aligned}$$

which under mild assumptions is **consistent** if $n \rightarrow \infty$ provided that $P(x_j = x_0) > 0$. **Only assuming we have many observations at x_0** . Consequently, this estimator is **not valid** if x_j is a continuous random variable. If data is continuous, $P(x = x_0) = 0$, integral at a point has no mass $\rightarrow 0 \rightarrow$ not valid.

7.1.5 Kernel regression : continuous x_i

For **continuous random variables** if $m(\cdot)$ is sufficiently smooth, we can estimate $m(x_0)$ using observations in a **neighborhood of x_0** . A **smoothing estimator** of $m(x_0)$ can be written as

$$\hat{m}(x_0) = \sum_{j=1}^n \omega_h(x_0 - x_j) y_j$$

where the weights $\omega_h(x_0 - x_j)$ **decreases** with the distance between x_j and x_0 . Frequently, the weights are defined as

$$\omega_h(x_0 - x_j) = \frac{K\left(\frac{x_0 - x_j}{h}\right)}{\sum_{k=1}^n K\left(\frac{x_0 - x_k}{h}\right)}$$

This is the **Nadaraya-Watson estimator** of $m(x_0)$. Under the regularity conditions, $\hat{m}(x_0)$ is **consistent** when $\lim_{n \rightarrow \infty} h = 0$, but convergence is slow.

The properties of the estimator depend on the choice of $K(\cdot)$ and h . The choice of $K(\cdot)$ is not that important, often just use $\phi(\cdot)$. But the **choice for h is critical**. If h is too small, the estimates become too irregular. If h is

too large, the estimates approach $\bar{y}$. A popular method to choose h is **leave-one-out cross validation**.

$$h_{cv} = \arg \min_h \frac{1}{n} \sum_{i=1}^n \delta(x_i) \left[y_i - \sum_{j \neq i} \omega_h(x_i - x_j) y_j \right]^2$$

where $\delta(x)$ is a **trimming function** to reduce boundary effects. $\delta(x)$ **trimming to leave out the tails, 0 in the tail** → **indicator function**. This can be generalized for a vector of regressors. However, for the multivariate case, not only do we have **curse of dimensionality** but also lose the ability to display results graphically.

7.2 Semiparametric estimation

Most non-ML estimators are semiparametric in the sense that they do not require the full specification of the conditional probability distribution of the dependent variable.

7.2.1 Partially linear model

Suppose $E[y_i|z, x] = z_i\beta + g(x_i)$ where $g(\cdot)$ is **left unspecified**, and

$$y_i = z_i\beta + g(x_i) + \epsilon_i, \quad i = 1, \dots, n$$

Robinson (1988) and Speckman (1988) have shown that under certain conditions it is possible to obtain $\sqrt{n}$ -consistent estimators for β . Notice we can write

$$\begin{aligned} E[y_i|x] &= E[z_i|x]\beta + g(x_i) \\ y_i - E[y_i|x] &= [z_i - E[z_i|x]]\beta + \epsilon_i \end{aligned}$$

证明. We begin with initial condition: $E[\epsilon_i|z_i, x_i] = 0$

$$\begin{aligned} E[y_i|x_i] &= E[z_i\beta + g(x_i) + \epsilon_i|x_i] \\ &= E[z_i\beta|x_i] + E[g(x_i)|x_i] + \underbrace{E[\epsilon_i|x_i]}_{(1)} = \boxtimes \\ (1)E[\epsilon_i|x_i] &= E[E[\epsilon_i|x_i, z_i]|x_i] = E[0|x_i] = 0 \\ \boxtimes &= \beta E[z_i|x_i] + g(x_i) \end{aligned}$$

Now moving on to the second equation,

$$\begin{aligned} y_i - \underbrace{E[y_i|x_i]}_{\text{Nadaraya-Watson}} &= z_i\beta + g(x_i) + \epsilon_i - E[z_i|x_i]\beta - g(x_i) \\ &= [z_i - \underbrace{E[z_i|x_i]}_{\text{Nadaraya-Watson}}]\beta + \epsilon_i \end{aligned}$$

So we have,

$$y_i - \widehat{E[y_i|x_i]} = [z_i - \widehat{E[z_i|x_i]}]\beta + \epsilon_i$$

□

If $E[y_i|x]$ and $E[z_i|x]$ are **known**, β can be estimated by **OLS**. However, if $E[y_i|x]$ and $E[z_i|x]$ are **replaced by kernel estimators** that converges sufficiently quickly, their use will not affect the rate of convergence of the **OLS** estimator of β

$$\hat{\beta} = \left[(Z - \widehat{E[Z|x]})'(Z - \widehat{E[Z|x]}) \right]^{-1} (Z - \widehat{E[Z|x]})'(Y - \widehat{E[Y|x]})$$

This estimator is **consistent and asymptotically normal**.

第 8 章 The Bootstrap

The bootstrap is a method introduced by Efron (1979) to estimate the distribution of an estimator, its variance, or the distribution of a test statistic by resampling the data. Suppose we have a sample $(X_1, \dots, X_n)$ and we would like to construct a confidence interval for a parameter θ . We estimate θ using a mathematical formula that depends on the sample $\hat{\theta} = T(X_1, \dots, X_n)$, $T(\cdot)$ is a known function. Let us assume that $\hat{\theta}$ is root-n consistent and $\hat{\theta}$ is asymptotically normal. That is $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \lambda^2)$. The **statistics of interest** are

- the estimator $\hat{\theta}$
- the standard errors of $\hat{\theta}$: $s_{\hat{\theta}}$ a consistent estimator of $\frac{\lambda}{\sqrt{n}}$
- the t-statistic $\hat{t} = \frac{(\hat{\theta} - \theta_0)}{s_{\hat{\theta}}}$ where $H_0 : \theta = \theta_0$
- the associated critical value or p-value for the test
- confidence interval

8.1 Basics on bootstrap

Let $(X_1^*, \dots, X_n^*)$ be a bootstrap sample generated from the original sample and $\hat{\theta}^* = T(X_1^*, \dots, X_n^*)$. Generate B different bootstrap samples and for each sample, $b = 1, \dots, B$, compute $\hat{\theta}^* = T(X_{1,b}^*, \dots, X_{n,b}^*)$.

8.1.1 Bootstrap percentile method

We can use the empirical distribution of $\hat{\theta}^*$ to approximate the asymptotic distribution of $\hat{\theta}$. For instance, the $100 \times (1 - \alpha)\%$ **confidence interval for θ based on** $(\hat{\theta}_{\frac{\alpha}{2}}^*, \hat{\theta}_{1-\frac{\alpha}{2}}^*)$, where $\hat{\theta}_{\alpha}^*$ is the empirical percentile $\alpha \times 100$ of $\hat{\theta}_b^*$. This is called the **bootstrap percentile method**.

In the **i.i.d. bootstrap method** proposed by Efron (1979), the bootstrap sample $(X_1^*, \dots, X_n^*)$ is obtained by drawing **samples with replacement** from the original sample $(X_1, \dots, X_n)$, where the probability of obtaining each X_i , $i = 1, \dots, n$, is $\frac{1}{n}$. **Denote $\bar{X}$ the average of the sample**. Bickel and Freedman (1981) and Singh (1981) proved that if the elements of the **original sample were i.i.d.**, the bootstrap distribution $\sqrt{n}(\bar{X}^* - \bar{X})$ is close to the asymptotic distribution of $\sqrt{n}(\bar{X} - \mu)$ for large n . This result justifies the validity of the bootstrap percentile method to construct a $100 \times (1 - \alpha)\%$ confidence interval for the **mean**:

$$(\bar{X}_{\frac{\alpha}{2}}^*, \bar{X}_{1-\frac{\alpha}{2}}^*)$$

where $\bar{X}^*$ is the quantile $100 \times \alpha$ of $\bar{X}^*$. $\underbrace{(\hat{\theta} - Z_{\frac{\alpha}{2}} SE(\hat{\theta}), \hat{\theta} + Z_{\frac{\alpha}{2}} SE(\hat{\theta}))}_{\text{asymptotically derived}} \approx \underbrace{(\hat{\theta}_{\frac{\alpha}{2}}^*, \hat{\theta}_{1-\frac{\alpha}{2}}^*)}_{\text{bootstrapping}}.$

We can use it to test hypothesis **if the alternative is two-sided**: $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$ at $100 \times \alpha\%$ level. **Rejection rule**: We reject H_0 in favor of H_1 if $\theta_0 \notin (\hat{\theta}_{\frac{\alpha}{2}}^*, \hat{\theta}_{1-\frac{\alpha}{2}}^*)$. There is no guarantee that the bootstrap percentile method performs better than the one obtained using asymptotic theory.

8.1.2 Bootstrap standard-errors

The bootstrap estimator of θ is $\bar{\theta}_B^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}_b^*$. The $s_{\hat{\theta}}^2$ can be approximated using

$$s_{boot}^2 = \frac{1}{B-1} \sum_{b=1}^B \left(\hat{\theta}_b^* - \bar{\theta}_B^* \right)^2 \quad (41)$$

Although more straight forward to compute than $s_{\hat{\theta}}^2$, there is not much work showing **consistency** of this in general setting. Even when consistent, there is no guarantee that this performs better than $s_{\hat{\theta}}^2$. Hahn and Liao (2021) showed that $n \times s_{boot}^2$ is **never smaller** than λ^2 as $n \rightarrow \infty$. Consequently, if one uses the statistic $\frac{(\hat{\theta} - \theta)}{s_{boot}}$ to make inference, it will **produce large confidence intervals and tests with size smaller than α** → the tests will not reject H_0 very often. **Inference will be incorrect and should avoid using s_{boot} .**

8.2 Bootstrap with asymptotic refinement

Asymptotic refinement refers to the ability of the bootstrap to provide approximations to the distributions of statistics that are more accurate than the approximations of conventional asymptotic distribution theory. This implies that the actual size of the tests based on bootstrap converges faster to α than the actual size of the tests based on the conventional asymptotic distribution theory. $P(N(0, 1) > Z_\alpha) = \alpha \Rightarrow P(|\hat{t}| > Z_{\frac{\alpha}{2}}) = \alpha + \frac{C_N}{\sqrt{N}}$ **using asymptotic theory, only approximates to $N(0, 1)$, converges asymptotically in distribution.** We need to **assume** that the statistic of interest is a **smooth function** of the data and that it is **asymptotically pivotal**.

定义 8.2.1. An **asymptotically pivotal statistic** means that the limit distribution does not depend on unknown parameters.

Most test statistics are asymptotically pivotal but most estimators are not.

Example

$H_0 : \theta = \theta_0, \sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \lambda^2)$. This is not asymptotically pivotal as the distribution depends on λ .

$H_0 : \theta = \theta_0, t = \frac{(\hat{\theta} - \theta_0)}{s_{\hat{\theta}}} \xrightarrow{d} N(0, 1)$. This is asymptotically pivotal.

We first generate B different bootstrap samples and for each sample, $b = 1, \dots, B$, compute $\hat{t}_b = \frac{(\hat{\theta}_b^* - \hat{\theta})}{s_{\hat{\theta}_b^*}}$. Note that we still need to obtain $s_{\hat{\theta}_b^*}^*$ which is the standard error computed with the bootstrap sample.

8.2.1 One sided test for t-statistic

Consider $H_0 : \theta = \theta_0$ vs $H_1 : \theta > \theta_0$. We define the bootstrap critical value $cv_{boot}(1 - \alpha)$ as the percentile $100 \times (1 - \alpha)$ of $(\hat{t}_1, \dots, \hat{t}_B)$. **Rejection rule:** We reject H_0 in favor of H_1 if $\hat{t}_{obs} > cv_{boot}(1 - \alpha)$, where $\hat{t}_{obs}$ is the observed value of the statistic in the sample. The actual size of the test is $\alpha + O(n^{-1})$ which is **better than $\alpha + O(n^{-1/2})$** .

8.2.2 Two sided test for t-statistic

Consider $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$. We have two ways to define the bootstrap critical values.

1. **non-symmetric test:** obtain the bootstrap critical value $cv_{boot}(\frac{\alpha}{2})$ and $cv_{boot}(1 - \frac{\alpha}{2})$ as the percentiles $100 \times \frac{\alpha}{2}$ and $100 \times (1 - \frac{\alpha}{2})$ of $(\hat{t}_1, \dots, \hat{t}_B)$ respectively. **Rejection rule:** We reject H_0 in favor of H_1 if $\hat{t}_{obs} \notin (cv_{boot}(\frac{\alpha}{2}), cv_{boot}(1 - \frac{\alpha}{2}))$. The actual size of the test is $\alpha + O(n^{-1})$. **Bootstrap doesn't perform better than using asymptotic theory.**
2. **symmetric test:** obtain the bootstrap critical value $cv_{boot}(1 - \alpha)$ as the percentile $100 \times (1 - \alpha)$ of $(|\hat{t}_1|, \dots, |\hat{t}_B|)$. **Rejection rule:** We reject H_0 in favor of H_1 if $|\hat{t}_{obs}| > cv_{boot}^a(1 - \alpha)$. The actual size of the test is $\alpha + O(n^{-3/2})$ which is **better than asymptotic theory**.

8.2.3 p-values

Consider $H_0 : \theta = \theta_0$ vs $H_1 : \theta > \theta_0$. The p-value is defined as $p = \frac{\sum_{b=1}^B 1(\hat{t}_b > \hat{t}_{obs})}{B}$. Consider $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$. The p-value is defined as $p = \frac{\sum_{b=1}^B 1(|\hat{t}_b| > |\hat{t}_{obs}|)}{B}$. In both cases, **rejection rule** is we reject H_0 in favor of H_1 if $p < \alpha$.

8.2.4 confidence interval

We can construct $100 \times (1 - \alpha)\%$ **confidence interval** using bootstrap t-statistic

- $(\hat{\theta} - s_{\hat{\theta}}cv_{boot}(1 - \frac{\alpha}{2}), \hat{\theta} - s_{\hat{\theta}}cv_{boot}(\frac{\alpha}{2}))$
- $(\hat{\theta} - s_{\hat{\theta}}cv_{boot}^a(1 - \alpha), \hat{\theta} + s_{\hat{\theta}}cv_{boot}^a(1 - \alpha)) \rightarrow \text{better}$

8.3 Bootstrap in the linear regression model

Consider the linear regression model $y_i = x_i'\beta + u_i$, $E[u_i|x_i] = 0$. Let $\hat{\beta}$ denote the **OLS** estimate of β and define the **residuals** as $\hat{u}_i = y_i - x_i'\hat{\beta}$.

8.3.1 Residual bootstrap

Denoting by u_i^* the **bootstrap errors**, generate the dependent variable as $y_i^* = x_i'\hat{\beta} + u_i^*$, $i = 1, \dots, n$.

- **Parametric residual bootstrap:** if the distribution of the errors is known and independent of the regressors, bootstrap errors can be generated from it.

- **Semiparametric residual bootstrap**: if the errors are iid and independent of the regressors, bootstrap errors can be obtained as draws with replacement from the vector of estimation residuals $\hat{u}_i$.

Independence between u_i and x_i is a must condition. Heteroskedasticity arises when $\text{Var}(u_i|x_i) = f(x_i)$, then cannot apply either of the methods.

8.3.2 Heteroskedasticity

- **Bootstrap of the pairs**: The **bootstrap observations** $(y_i^*, x_i'^*)$ can be obtained as draws with replacement from the sample $\{(y_1, x_1'), \dots, (y_n, x_n')\}$. We obtain n such observations. Draw the y's and x's together then can use maximum likelihood.
- **Wild-bootstrap**: A variant of semiparametric residual bootstrap. To preserve the link between the errors and the regressors, the bootstrap error for observation i can be obtained as $u_i^* = f(\hat{u}_i)\epsilon_i^b$, where
 - $f(u_i)$ is an appropriate function of the residuals, e.g. the identity function of $f(\hat{u}_i) = \hat{u}_i$. Normally just use the residuals.
 - ϵ_i^b is a random variable independent of u and x such that $E[\epsilon_i^b] = 0$ and $\text{Var}(\epsilon_i^b) = 1$. We can use the **Rademacher distribution** for ϵ_i^b where $F_1 : \epsilon_i^b = \begin{cases} -1 & \text{with possibility } \frac{1}{2} \\ 1 & \text{with possibility } \frac{1}{2} \end{cases}$

For this method we can only use OLS estimation.

8.3.3 Heteroskedasticity and serial correlation

Moving blocks bootstrap: In the case the data are serially correlated, we draw **overlapping blocks** of ℓ consecutive observations from the sample $\{(y_1, x_1'), \dots, (y_n, x_n')\}$ with replacement until we have a sample of size n . When the observations are correlated, it doesn't make sense to draw individually, you want to keep the correlation. The issue is more with choosing the size of the block.

8.4 Number of bootstraps

For iid bootstrap and a sample of size n there are n^n possible bootstrap samples of size n . For **standard error** computation, Efron and Tibsharan (1993) say $B = 50$ is often enough, and $B = 200$ is almost always enough. For hypothesis testing and confidence intervals at standard levels of statistical significance involve the tails of the distribution, so more repetitions are needed. For **hypothesis testing** at level choose B so that $(B+1)\alpha$ is an integer. Davidson and MacKinnon (1999, 2000) recommend $B \geq 399$ if $\alpha = 0.05$ and $B = 1499$ if $\alpha = 0.01$. In the case of **inference** on means using **t-statistics** and iid sample, Babu and Singh (1983) suggest $B = n(\log n)^2$.

8.5 Final remarks

Although the bootstrap is often accurate, it is not always valid. There are several examples where bootstrap leads to **invalid inferences**.

Example

- Distributions with heavy tails
- Inference about a parameter that is on the boundary of the parameter set → inference on σ^2 in $[0, \infty)$
- Inference about the maximum or minimum of random variables

If the quantities involved are **not asymptotically pivotal**, the bootstrap procedures are not better than usual methods based on asymptotic theory but may be easier to implement.

第 9 章 Panel Data Models

We can follow the same random individual observations over time, known as **panel data** or longitudinal data. In general, we have n units each of whom we follow for T_i consecutive time periods, for each unit $i = 1, \dots, n$. The size of the sample that matters here for asymptotic theory to work is n , not T or $n \times T$.

Advantages include 1) controlling for individual unobserved heterogeneity and 2) identify individual-specific and time effects. Limitations include 1) design and data collection problems and 2) usually short time-series dimension.

9.1 Models

9.1.1 Pooled regression

This is the **most restrictive model** that specifies constant coefficients

$$y_{it} = x'_{it}\beta + \epsilon_{it} \quad (42)$$

where $\epsilon_{it} \sim IID(0, \sigma_\epsilon^2)$.

OLS gives **unbiased, consistent and most efficient** estimates of β if x_{it} is independent of ϵ_{it} → **Gauss-Markov Theorem**. The model **does not allow** explicitly for **individual heterogeneity**.

9.1.2 Fixed effects model

The model is defined as

$$y_{it} = \alpha_i + x'_{it}\beta + \epsilon_{it} \quad (43)$$

where $\epsilon_{it} \sim IID(0, \sigma_\epsilon^2)$ and α_i capture unobserved individual heterogeneity. If we ignore α_i → **correlation between x_{it} and ϵ_{it} → inconsistency with the OLS estimator as $u_{it} = \alpha_i + \epsilon_{it}$** . The estimator for β is **consistent and unbiased** even if we consider the variables **α_i either as fixed parameters or as random variables** and **correlated with x_{it} but independent of ϵ_{it}** . We require **the regressors to vary with t** . The coefficients of the non-varying regressors are not identified.

9.1.3 Random effects model

The model is defined as

$$y_{it} = \alpha_i + x'_{it}\beta + \epsilon_{it} \quad (44)$$

where

$$\epsilon_{it} \sim IID(0, \sigma_\epsilon^2) \quad \alpha_i \sim IID(0, \sigma_\alpha^2)$$

where α_i captures individual unobserved heterogeneity but are **not correlated with x_{it}** . $\alpha_i + \epsilon_{it}$ are treated as an error consisting of two components. This has a stronger assumption than FE but estimators are more efficient than FE.

9.1.4 FE vs. RE

- If there are **no omitted variables** or **omitted uncorrelated with explanatory** → use **RE**.
- If there **are omitted** and probability **correlated with explanatory** → use **FE**. But the omitted must have time-invariant values with time-invariant effects.
- If key explanatory is **time invariant** → must use **RE**.

9.2 FE estimators

9.2.1 Within estimator / fixed effects estimator

Relies on **within transformation** by defining the averages over t for y_{it} , x_{it} and ϵ_{it} and we get

$$\begin{aligned} y_{it} - \bar{y}_i &= (x_{it} - \bar{x}_i)' \beta + (\epsilon_{it} - \bar{\epsilon}_i) \\ &\Downarrow \\ \ddot{y}_i &= \ddot{X}_i' \beta + (\epsilon_{it} - \bar{\epsilon}_i) \end{aligned}$$

then use **OLS** on the transformed model with

$$\hat{\beta}_{within} = \left[\sum_{i=1}^n \ddot{X}_i' \ddot{X}_i \right]^{-1} \times \left[\sum_{i=1}^n \ddot{X}_i' \ddot{y}_i \right] = [W_{xx}]^{-1} W_{xy}$$

- **Unbiased**: ϵ_{it} independent of x_{is} for all t and s . $E[\hat{\beta}_{within}] = \beta \rightarrow$ independence.
- **Consistent**: as $n \rightarrow \infty$, $E[(x_{it} - \bar{x}_i)\epsilon_{it}] = 0$. Sufficient to have $E[x_{it}\epsilon_{it}] = 0$ and $E[\bar{x}_i\epsilon_{it}] = 0$ which is implied by the unbiasedness condition. In this case we call x_{is} **strictly exogenous**.

定义 9.2.1. *Strictly exogenous variables are not allowed to depend upon current, future, and past values of the error term. Strict exogeneity excludes inclusion of past values of y_{it} in x_{it} .*

- **Asymptotically efficient**: ϵ_{it} is homoskedastic with variance σ_ϵ^2
 - $\sqrt{n}(\hat{\beta}_{within} - \beta_0) \xrightarrow{d} \mathcal{N}(0, \text{avar}(\hat{\beta}_{within}))$
 - if under heteroskedasticity and serial correlation by clusters then use **robust SE** proposed by Arelano (1987) which is good for small $\frac{T}{n}$.

9.2.2 Least square dummy variable estimator (LSDV)

Assume α_i are parameters to be estimated. Rewrite the model as

$$y_{it} = \sum_{j=1}^n \alpha_j d_{ij} + x'_{it} \beta + \epsilon_{it}$$

where $d_{ij} = 1$ if $i = j$, 0 otherwise. The parameters $\{\alpha_1, \dots, \alpha_n, \beta\}$ are estimated using **OLS**. The estimator obtained for β using this approach is the same as $\hat{\beta}_{within}$.

Considering $\hat{\alpha}_i = \bar{y}_i - \bar{x}'_i \hat{\beta}_{within}$ $i = 1, \dots, n$

- **Consistent**: requires $T \rightarrow \infty$. As $n \rightarrow \infty$ while T fixed, $\hat{\alpha}_i$ does not converge. If we want $\hat{\alpha}_i$ to be consistent as well, then we need both n and T to go to infinity.

9.2.3 First difference estimator (FD)

By subtracting one period from the other, we obtain the **first difference equation**

$$\Delta y_{i,t} = \beta' \Delta x_{i,t} + \Delta \epsilon_i$$

where $\Delta a_{i,t} = a_{it} - a_{i,t-1}$. Then use **OLS** to get

$$\hat{\beta}_{FD} = \left[\sum_{i=1}^n \sum_{t=1}^T (\Delta x_{it})(\Delta x_{it})' \right]^{-1} \times \left[\sum_{i=1}^n \sum_{t=1}^T (\Delta x_{it})(\Delta y_{it}) \right]$$

9.2.4 FE vs. FD

- When $T = 2$, FE and FD are identical $\rightarrow$ use **FD** because it's more straightforward and easier to compute heteroskedasticity-robust statistics.
- When $T \geq 3$, both unbiased and consistent for large n and small T , depends on the relative efficiency which depends on heteroskedasticity and serial correlation in the idiosyncratic errors.
 - if ϵ_{it} **uncorrelated and homoskedastic** $\rightarrow$ **within**
 - if ϵ_{it} **substantially autocorrelated**, e.g. RW, $\Delta \epsilon_{it}$ uncorrelated $\rightarrow$ **FD**
 - if ϵ_{it} has **some serial autocorrelation** but not as much as RW, then **cannot easily compare**.

9.3 RE estimators

9.3.1 Pooled OLS

Treats individual-specific effects α_i as random factors iid over individuals. This estimator is **consistent but inefficient** because it ignores the correlation among the $\omega_{it} = \alpha_i + \epsilon_{it}$. All valid inference requires the

robust covariance estimator.

α_i and ϵ_{it} are assumed to be **mutually independent** and **independent** of x_{js} for all j and s . For each individual, define $\iota_T = (1, \dots, 1)'$ of dimension $T \times 1$. The model can be written as

$$y_i = X_i \beta + \omega_i \quad i = 1, \dots, n$$

where $\omega_i = \alpha_i \iota_T + \epsilon_i$ and y_i and X_i are matrix of $T \times 1$ dimension. For each individual, we have

$$\text{Var}(\omega_i) = \Sigma = \sigma_\alpha^2 \iota_T \iota_T' + \sigma_\epsilon^2 I_T$$

The model can once again be rewritten as

$$Y = X\beta + \omega$$

where all three variables are $N \times 1$ dimension matrix and we have $E[\omega] = 0$ and $\text{Var}(\omega) = \Psi = I_n \otimes \Omega$.

9.3.2 Between estimator

This alternative estimator is **unbiased and consistent but not efficient**. Run the regression on

$$\begin{aligned} \bar{y}_i &= \bar{x}_i' \beta + \alpha_i + \bar{\epsilon}_i \\ &\Downarrow \\ \hat{\beta}_B &= \left[\sum_{i=1}^n \bar{x}_i \bar{x}_i' \right]^{-1} \sum_{i=1}^n \bar{x}_i \bar{y}_i = [B_{xx}]^{-1} B_{xy} \end{aligned}$$

9.3.3 Generalized least squares (GLS)

The model in **Pooled OLS** does not satisfy the Gauss-Markov assumptions. Thus we can transform the model into one that satisfies the model by finding a nonsingular P such that $\Psi^{-1} = P'P$.

$$\begin{aligned} Y &= X\beta + \omega \\ E[\epsilon|X] &= 0 \quad \text{Var}[\epsilon|X] = \Psi \\ &\Downarrow \\ PY &= PX\beta + P\epsilon \\ E[P\epsilon|X] &= PE[\epsilon|X] = 0 \quad \text{Var}(P\epsilon|X) = P\Psi P' = I \\ &\Downarrow \\ Y^* &= X^*\beta + \epsilon^* \end{aligned}$$

The model can be estimated by **OLS** to give the **BLUE**, best linear unbiased estimator, for $\beta \rightarrow \hat{\beta}_{GLS}$

$$\hat{\beta}_{GLS} = (X'^* X^*)^{-1} X'^* Y^* = (X' \Psi^{-1} X)^{-1} X' \Psi^{-1} Y$$

Typically Ψ is unknown and need to be estimated in the first step.

To **find the expression for P** , one can show that $P = I_n \otimes \Sigma$ where $\Sigma = [I_T - \frac{\theta}{T} \iota_T \iota_T' / \sigma_\epsilon]$ where $\theta = 1 - \frac{\sigma_\epsilon^2}{\sigma_\epsilon^2 + T\sigma_\alpha^2}$. Thus we have

$$\hat{\beta}_{GLS} = \left(\sum_{i=1}^n X_i' \Omega^{-1} X_i \right)^{-1} \sum_{i=1}^n X_i' \Omega^{-1} Y_i$$

run regression on $\Downarrow y_{it} - \theta \bar{y}_i = (x_{it} - \theta \bar{x}_i)' \beta + (\eta_{it} - \theta \bar{\eta}_i)$

$$\hat{\beta}_{GLS} = \left[\sum_{i=1}^n \sum_{t=1}^T (x_{it} - \theta \bar{x}_i)(x_{it} - \theta \bar{x}_i)' \right]^{-1} \times \left[\sum_{i=1}^n \sum_{t=1}^T (x_{it} - \theta \bar{x}_i)(y_{it} - \theta \bar{y}_i) \right]$$

Combining the **FE within estimator** and the **RE between estimator**, we can rewrite the GLS estimator as

$$\hat{\beta}_{GLS} = [W_{xx} + \lambda B_{xx}]^{-1} [W_{xx} \hat{\beta}_{within} + \lambda \beta_{xx} \hat{\beta}_B]$$

$$\lambda = \frac{\sigma_\epsilon^2}{\sigma_\epsilon^2 + T\sigma_\alpha^2}$$

$$\Downarrow$$

$$\theta = 1 - \lambda^{1/2}$$

- if $\lambda = 0 \rightarrow$ fixed effect estimator $\hat{\beta}_{within}$. The GLS estimator is a linear combination of the within and between estimator.
- if $\lambda = 1 \rightarrow$ no individual specific effects $\alpha_i \rightarrow$ pooled OLS estimator. Homoskedasticity but no autocorrelation.

9.3.4 Feasible GLS

Considering the **transformed model**

$$(y_{it} - \theta \bar{y}_i) = (x_{it} - \theta \bar{x}_i)' \beta + \underbrace{(\eta_{it} - \theta \bar{\eta}_i)}_{u_{it}} \quad u_{it} \sim IID$$

We need to estimate θ and therefore λ .

$$\hat{\sigma}_\epsilon^2 = \frac{1}{n(T-1)} \sum_{i=1}^n \sum_{t=1}^T \left[y_{it} - \bar{y}_i - (x_{it} - \bar{x}_i)' \hat{\beta}_{within} \right]^2 \rightarrow \text{within group residuals}$$

$$\hat{\sigma}_B^2 = \frac{1}{n} \sum_{i=1}^n (\bar{y}_i - \bar{x}_i' \hat{\beta}_B)^2 \rightarrow \text{error variance from between}$$

$$\hat{\sigma}_\alpha^2 = \hat{\sigma}_B^2 - \frac{1}{T} \hat{\sigma}_\epsilon^2$$

↓

$$\hat{\theta} = 1 - \sqrt{\hat{\lambda}} \quad \hat{\lambda} = \frac{\hat{\sigma}_\epsilon^2}{\hat{\sigma}_\epsilon^2 + T \hat{\sigma}_\alpha^2}$$

Run regression ↓ $(y_{it} - \hat{\theta} \bar{y}_i) = (x_{it} - \hat{\theta} \bar{x}_i)' \beta + u_{it}$

$$\hat{\beta}_{RE} = \left[\sum_{i=1}^n \sum_{t=1}^T (x_{it} - \hat{\theta} \bar{x}_i)(x_{it} - \hat{\theta} \bar{x}_i)' \right]^{-1} \times \left[\sum_{i=1}^n \sum_{t=1}^T (x_{it} - \hat{\theta} \bar{x}_i)(y_{it} - \hat{\theta} \bar{y}_i) \right]$$

- if $\theta = 1 \rightarrow \hat{\beta}_{within}$
- if $\theta = 0 \rightarrow \hat{\beta}$ the Pooled OLS estimator

9.3.5 Choosing estimator in the RE set up

Given **Random Effect model**, which estimation method should we use? With **strict exogeneity of x_{it}**

- the **within estimator** is **consistent but not efficient** because it ignores the structure of the covariance matrix Ω .
- the **first difference estimator** is also **consistent but not efficient**
- the **between estimator** is **consistent but not efficient** because it does not take into account information from within dimension
- the **RE estimator** is **consistent and asymptotically efficient** under **strict exogeneity** assumption and **uncorrelated with α_i** .

All estimators are consistent but only the RE estimator is efficient, i.e., has the smallest variance.

The choice of estimator really depends on testing **whether α_i are correlated with the regressors or not** →

Hausman Test

$$H_0 : x_{it}, \alpha_i \text{ uncorrelated} \quad H_1 : x_{it}, \alpha_i \text{ correlated}$$

↓

$$H_0 : \hat{\beta}_{within} \text{ consistent} \quad \hat{\beta}_{RE} \text{ consistent and efficient}$$

$$H_1 : \hat{\beta}_{within} \text{ consistent} \quad \hat{\beta}_{RE} \text{ inconsistent}$$

Test statistic ζ_H is based on the difference $(\hat{\beta}_{within} - \hat{\beta}_{RE})$. Under H_0 , $\lim(\hat{\beta}_{within} - \hat{\beta}_{RE}) \xrightarrow{p} 0$ since both estimators are consistent. Under H_0 , $\zeta_H \xrightarrow{d} \chi^2(k)$ where k is the dimension of β . If the inverse of the variance matrix of the within and the RE estimator **cannot be computed**, just **use the same σ_e^2** both within and RE estimator.

第 10 章 GARCH Models